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Institut des Énergies

System Level Modelling and Simulation of MVDC Distribution Grids featuring Solid State Transformers

ICDCM 2026 - Tutorial

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Hes·so

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- Applied Research & Development (linking academy and industry)
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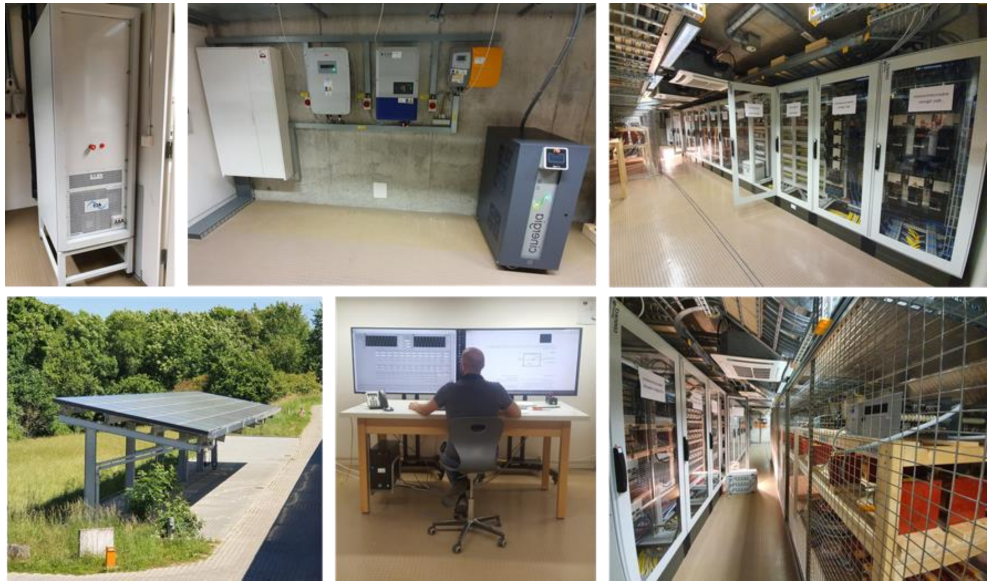

Former Affiliations

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KTH Vetenskap och Konst
CERN
ABB
Hitachi Energy

Tutorial Outline

Content

- ❑ Introduction
  - DC grid basics and standards
  - Modeling basics
- ❑ System level Models
  - Elementary DAB-based SST cell
  - DC bus, active front end and batteries, Series/parallel connection of SSTs
- ❑ Model of an MVDC microgrid
  - DC bus with multiple AFE/PFEs, Full model of an MVDC power system
  - Continuous run approach
- ❑ Conclusions and Prospects

- ❑ The tutorial will be composed half/half with traditional slides and practical demonstrations that the audience can follow and redo on their own computers.
- ❑ The models are built and demonstrated with *Powersys Simba**, a simple and powerful simulation environment that allows Python automated scripting and a so-called continuous time simulation approach.

**the presented models are easily implementable in other similar simulation software environments*

TUTORIAL CONTENT

Sharing models

Get the models

- ❑ https://github.com/PESC-CH/System-level-MVDC-with-SST

Or play online from Simba Cloud

https://simba.io/share/BOkZ3MTIOphfx2d0GO_r_5r0t1-WJD-FqdD6U_JANPuuzBj2vQq08uga4vFgK4WX/

INTRODUCTION

DC grid basics

Comparing DC to AC

- When compared to AC grids, similarities can be drawn:
  - Primary control is related to active power (no reactive), preferably implemented via droop control (also related to virtual resistance).
  - Secondary and Tertiary control are related to power and grid management, requiring communication between converter units or between grids.
  - A grid can be modeled with ideal sources and its Thevenin/Norton equivalent reflecting its impedance.

- When modeling DC grids for study there are still some open questions:
  - **Protection** is not as simple as in AC grid because there is no zero current crossing for easy breaking features
  - **What does short circuit capability mean in DC grids?** Is it related to converters' limits while operating DC transformers?
  - **What does inertia mean in DC grids?** In AC grids, inertia is related to the rotating momentum of generators. Is DC inertia related to stored energy in capacitors?

INTRODUCTION

Standards

Initiatives

- ❑ Lack of standardized protocols for components and systems integration
- ❑ some recommendations can be found
  - **Cigre** Technical Brochure: Medium Voltage DC Distribution Systems (TB875)
  - **Current/OS** list of recommendations
  - **IEC white paper** - Medium voltage DC (MVDC) grids for an all-electric society
  - **IEEE P3105**: Recommended Practice for Design and Integration of Solid State Transformers in Electric Grid *(in work)*

Standards

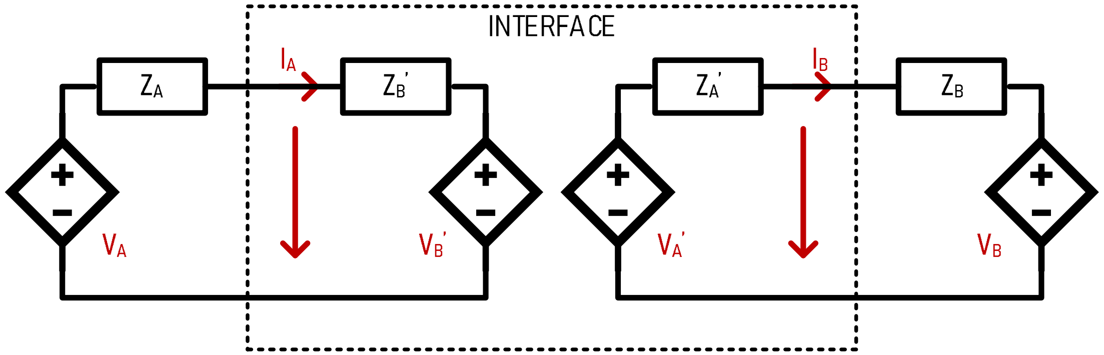
- ❑ IEEE Std 1709-2018
  - This document is IEEE Recommended Practice for 1 kV to 35 kV MVDC power systems on ships.
- ❑ IEEE 2030.10-2021
  - This document is IEEE Standard for LVDC Microgrids for Rural and Remote Electricity Access Applications.
- ❑ IEC TR 63282:2020
  - LVDC systems -Assessment of standard voltages and power quality requirements

INTRODUCTION

Modeling basics

How to model

- The work behind the modeling that is presented in this tutorial is inspired by a common practice in Hardware-in-the-Loop Real Time Modelling where all systems and subsystems are defined through their Thevenin/Norton equivalent.



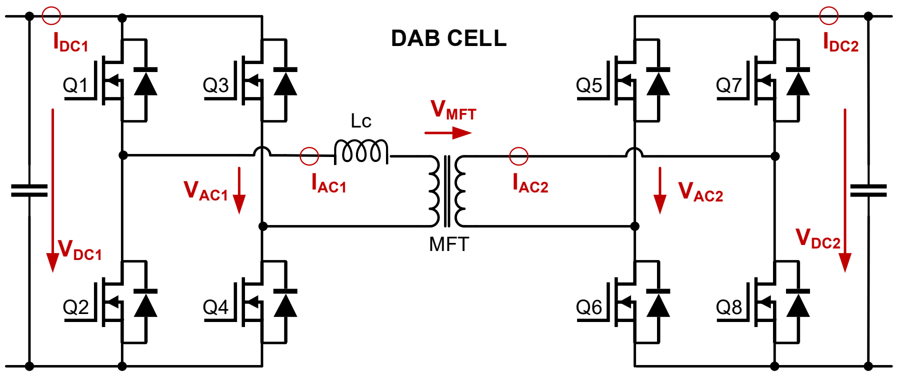
Standards

- IEEE P2004 (DRAFT) : Recommended Practice for Hardware-in-the-Loop (HIL) Simulation-Based Testing of Electric Power Apparatus and Controls
- This is interesting because it discusses the way to interface subsystems through equivalent source and equivalent impedance as seen from the other subsystem

System level Models

Elementary DAB-based SST cell

- Dual Active Bridge (DAB) is a promising candidate for acting as modular SST building bloc
- The active power flow is phase shift controlled through pulses applied to the Medium Frequency Transformer
- This study will implement a system level model of a converter that has been built and tested in laboratory
- The operating parameters and passive elements reflect its overall dynamics and behaviors



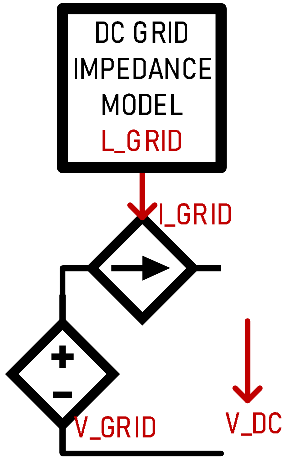
S. Heinig et D. Siemaszko et al., "Experimental Insights into the MW Range Dual Active Bridge with Silicon Carbide Devices," 2022 International Power Electronics Conference (IPEC-Himeji 2022-ECCE Asia), Himeji, Japan, 2022

	Symbol	Value	Comment
DC nominal voltage on primary side	V_{PRIM}	1000V	Set by grid model V_{GRID}
DC nominal voltage on secondary side	V_{SEC}	2000V	Current/Voltage control by SST
nominal current on secondary side	I_{SEC}	250A	Set by load model I_{LOAD}
maximum current on secondary side	I_{SEC_MAX}	300A	Voltage control Limiter by SST
MFT side SST inductance	L_{MFT}	8.5µH	Defines control's time response
DC side SST capacitor	C_{DC}	3mF	Defines control's time response
MFT switching frequency	F_{SW}	10kHz	Defines control's transfer function
Simulation time step	t_s	1µs	This time step is used in all runs

System level Models

Elementary DAB-based SST cell

- The DAB based SST is modeled with it's equivalent Thevenin/Norton equivalents.
- Voltage and current sources are controlled in a way it reflects its dynamics as seen from its terminals.
- The grid model is an ideal DC voltage source with a current source in series that implements the DC grid impedance.
- The impedance model is given through its equivalent RL values.



$$I_{GRID} = \frac{1}{L_{GRID}} \int (V_{DC} - V_{GRID} - R_{GRID} I_{GRID})$$

System level Models

Elementary DAB-based SST cell

- The MFT is modeled with two current sources (one per winding)
- The current is related to leakage inductor L_{MFT}, switching frequency F_{SW} and applied voltage V_{MOD}

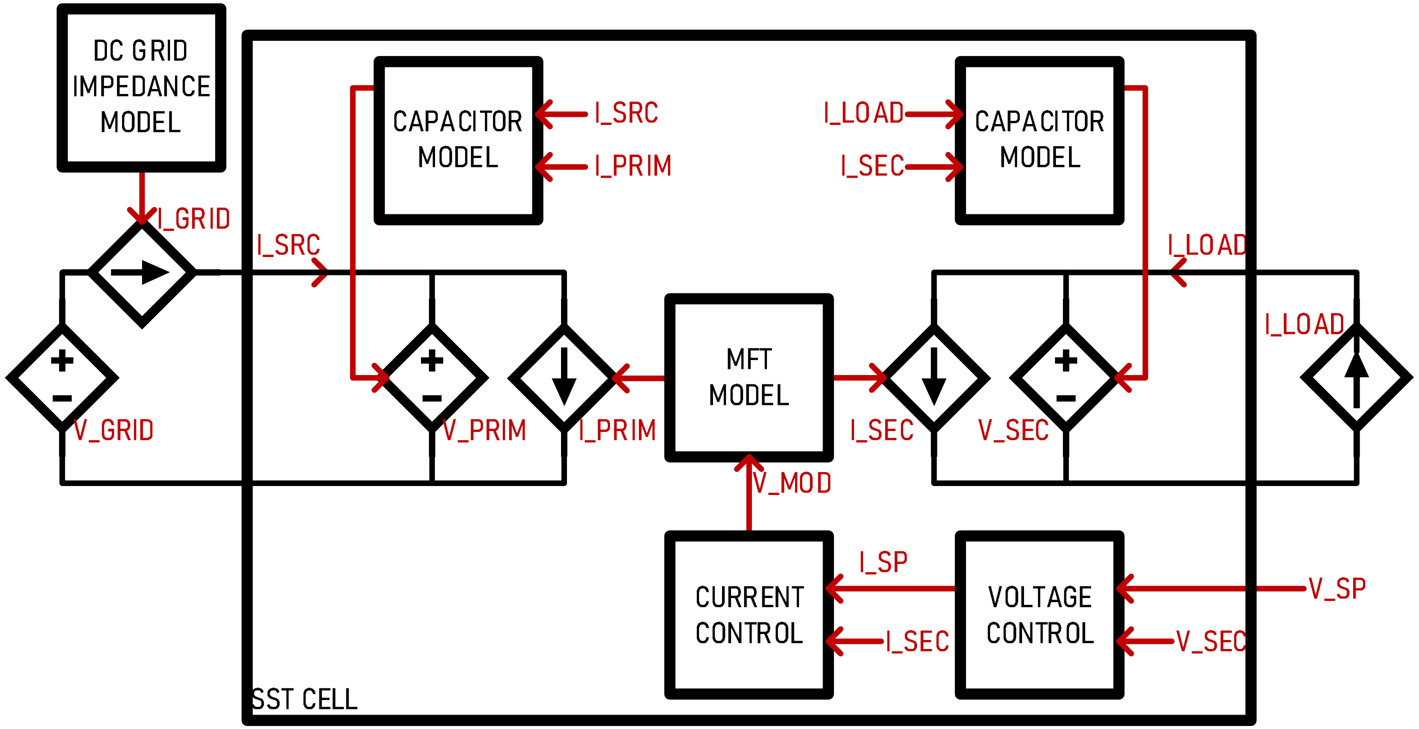
- Capacitors are modeled as voltage sources
- The voltage is related to difference in currents from interface and winding, and capacitor value C_{DC}

$$V_{PRIM} = \frac{1}{C_{DC}} \int (I_{SRC} - I_{PRIM})$$

$$V_{SEC} = \frac{1}{C_{DC}} \int (I_{LOAD} - I_{SEC})$$

$$I_{SEC} = \frac{1}{L_{MFT} F_{SW}} V_{MOD}$$

$$I_{PRIM} = I_{SEC} N_{MFT}$$



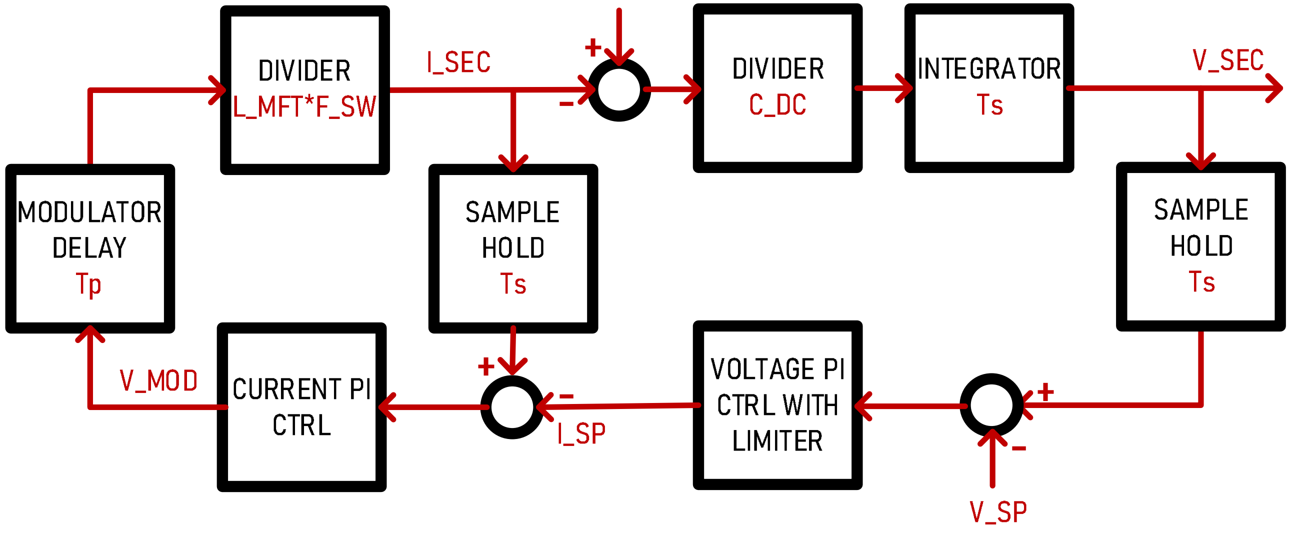
System level Models

Elementary DAB-based SST cell

- The modulation voltage V_{MOD} applied to the leakage inductor is given by a current/voltage cascaded PI controller
- The full system implements delays related to sampling frequency T_S and modulator's inherent delay due to switching period T_P

$$V_{MOD} = K_P(I_{SEC} - I_{SP}) + K_I \int (I_{SEC} - I_{SP})$$

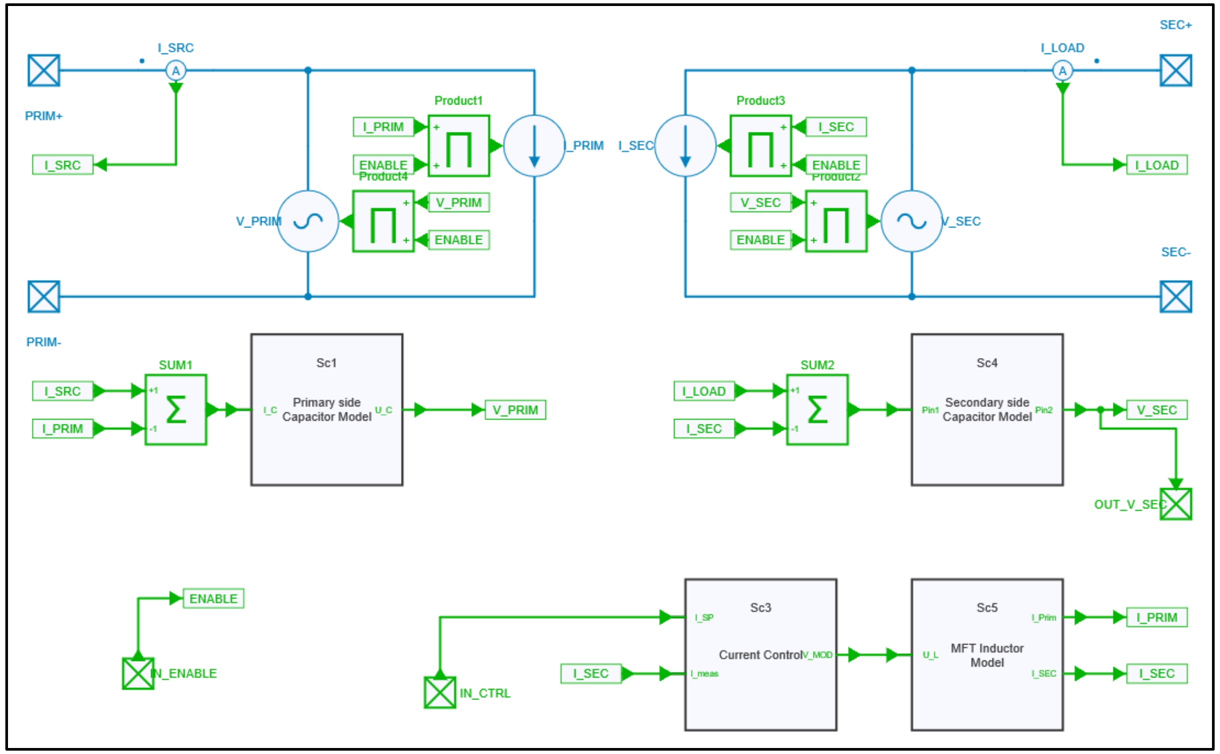
$$I_{SP} = K_P(V_{SEC} - V_{SP}) + K_I \int (V_{SEC} - V_{SP})$$

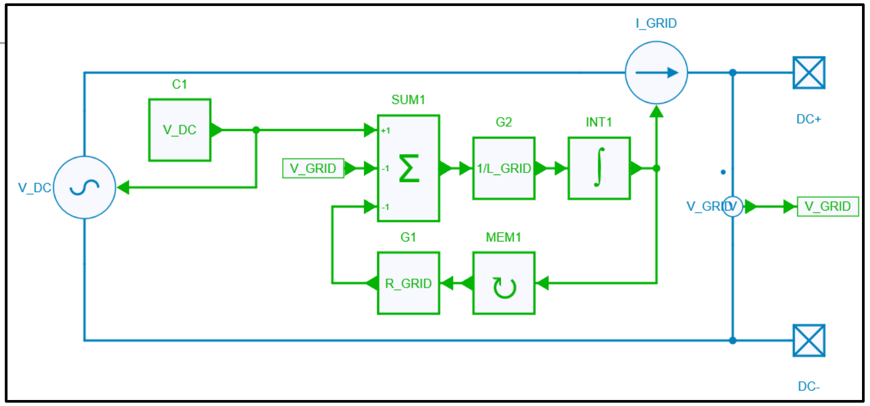


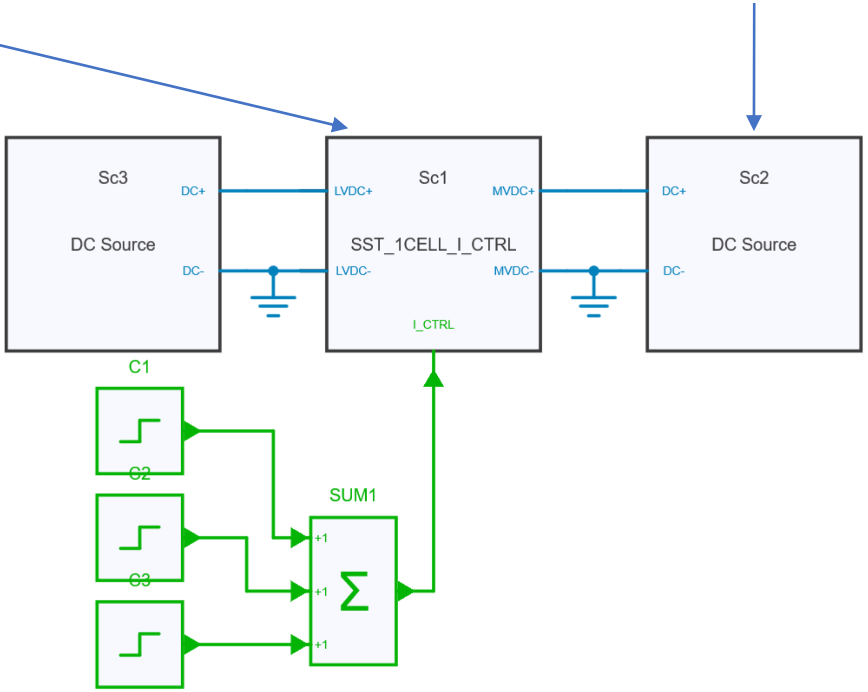
System level Models

Elementary DAB-based SST cell

- ❑ *Run model “1 Single SST Current CTRL” from file “SST_DCMicroGrid_Models.jsimba”*



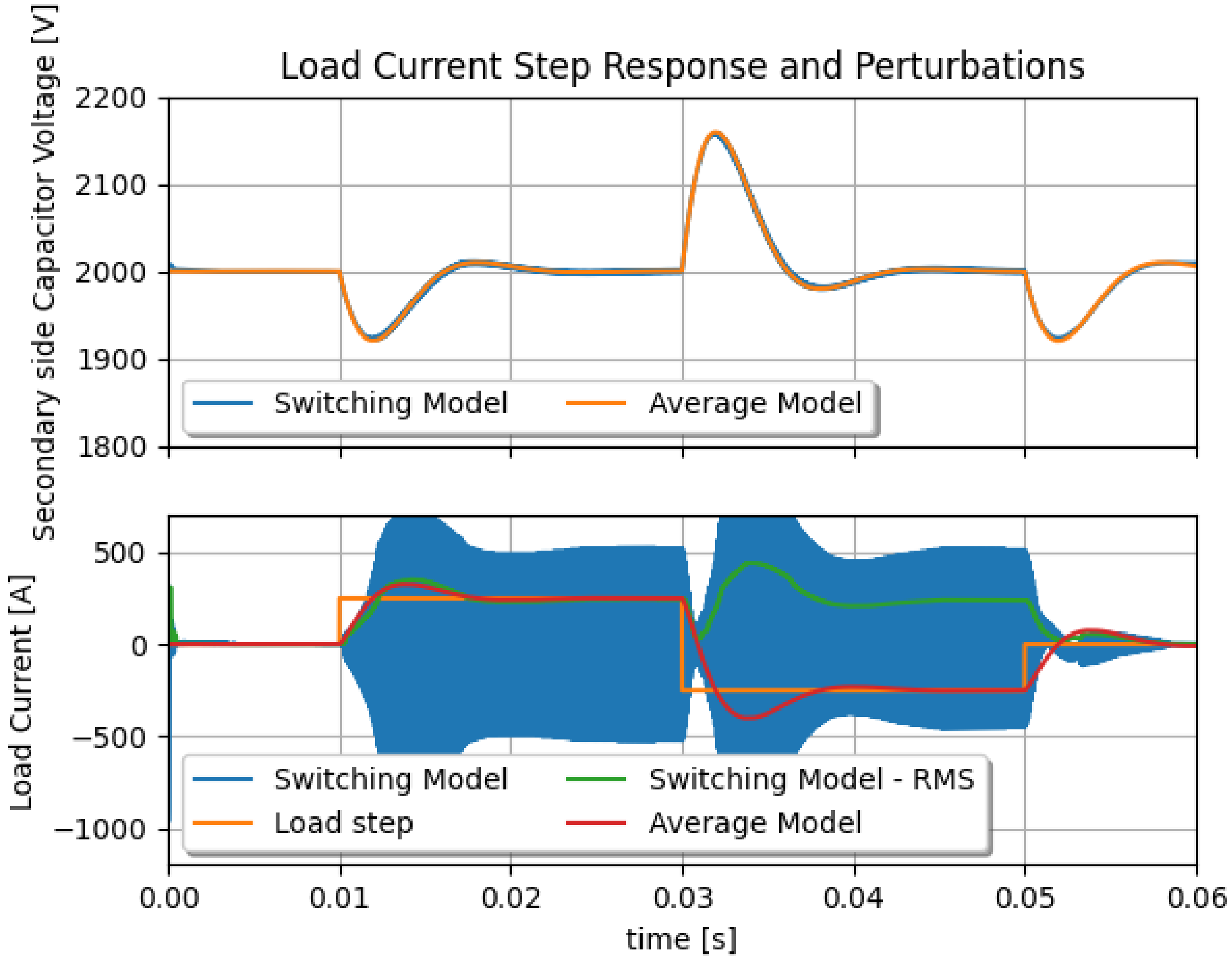




https://github.com/powelsys/System-level-MVDC-with-SST

SYSTEM LEVEL MODELS

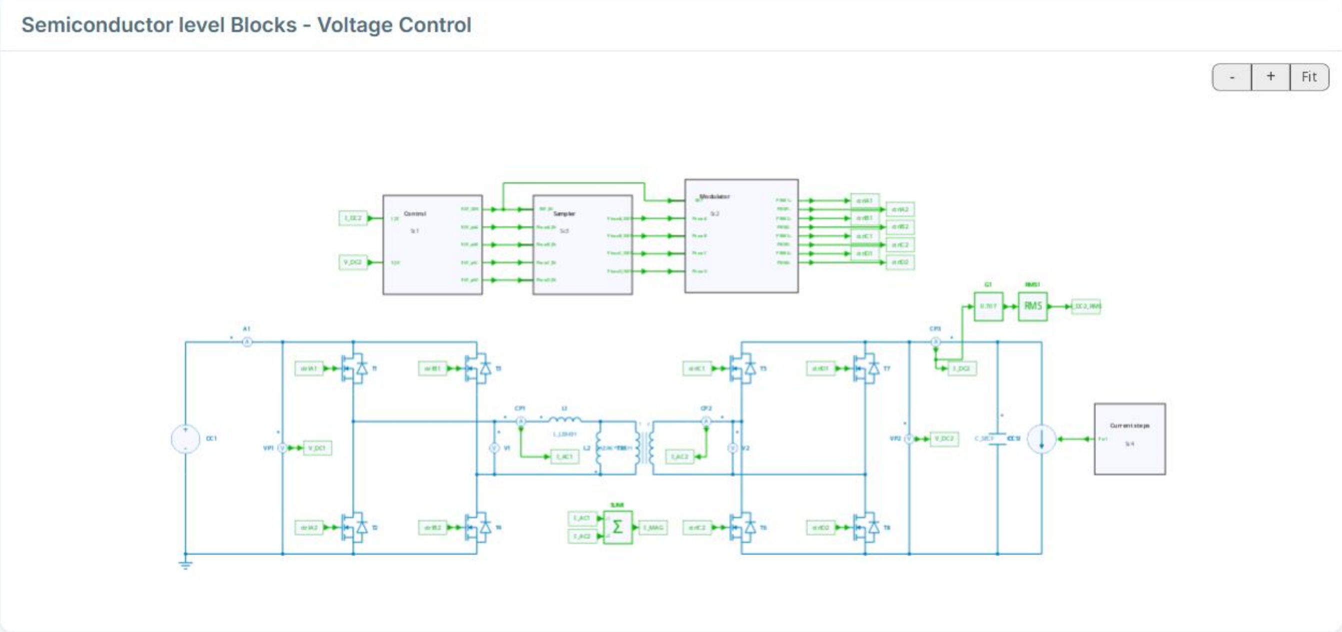
Elementary DAB-based SST cell



TUTORIAL CONTENT

Sharing models

❑ Dual Active Bridge - Isolated DC/DC converter



Circuit Variables

Name	Value
V_OUT_RI	2000
I_LOAD	250
V_IN_REF	1000
C_SEC	3E-3
L_LEAK	8.5e-6
KI_I	7500
KP_I	0.3
KI_V	1000.0
KP_V	2
I_SST_LIN	500.0

https://simba.io/share/c06cOm9JwnQmkMRzxVOx_gTBA9r9lGUPr8ykr6kFMaYFhikCkP12guffLvUJVL7T/

SYSTEM LEVEL MODELS

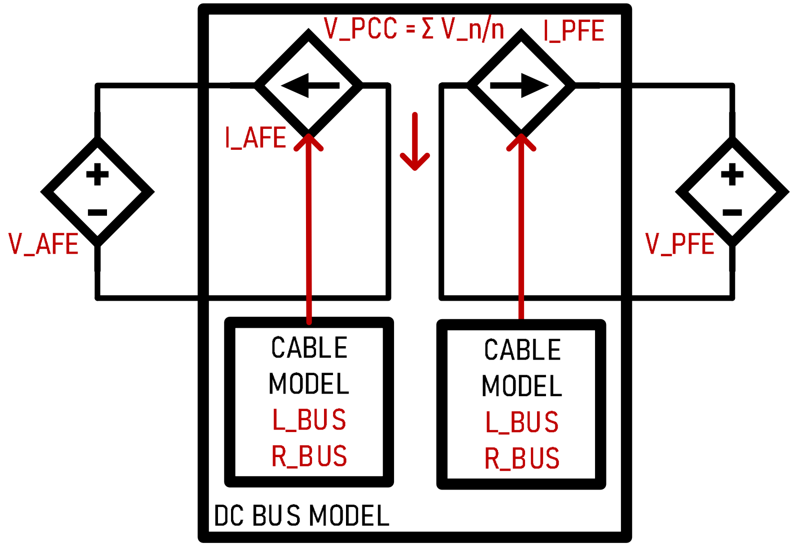
DC bus

- ❑ Connecting voltage sources can only be done through current sources
- ❑ In this case we use a cable RL model with resistance and inductor for generating the current between DC ports.
- ❑ The Voltage as seen at the Point of Common Coupling is the pondered mean value between all applied voltages
- ❑ We differentiate Active Front End (maintaining the voltage) and Passive Front End (only drawing currents)

$$V_{PCC} = \frac{V_{AFE}+V_{PFE}}{2}$$

$$I_{AFE} = \frac{1}{L_{DCBUS}} \int (V_{PCC} - V_{AFE} - R_{DCBUS} I_{AFE})$$

$$I_{PFE} = \frac{1}{L_{DCBUS}} \int (V_{PCC} - V_{PFE} - R_{DCBUS} I_{PFE})$$

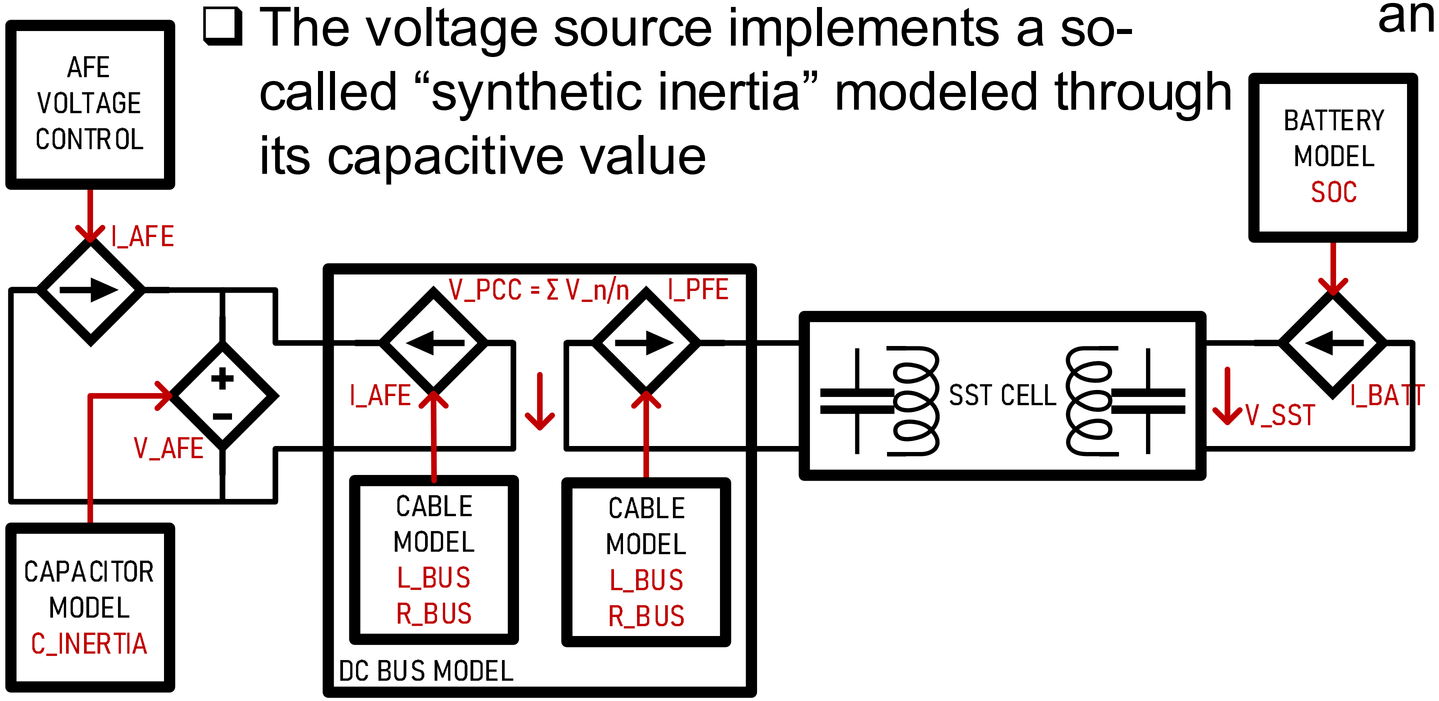


System level Models

Active Front End and Battery Storage

- The active front end is a current source feeding a voltage source
- The dynamics are related to a PI controller that reflects real hardware
- The voltage source implements a so-called "synthetic inertia" modeled through its capacitive value

- The Battery model implements the idea of State of Charge connected to a current source.
- The Battery can only operate between 10 and 90%



$$SoC = \int \frac{I_{BATT}}{Q_{BATT}}$$

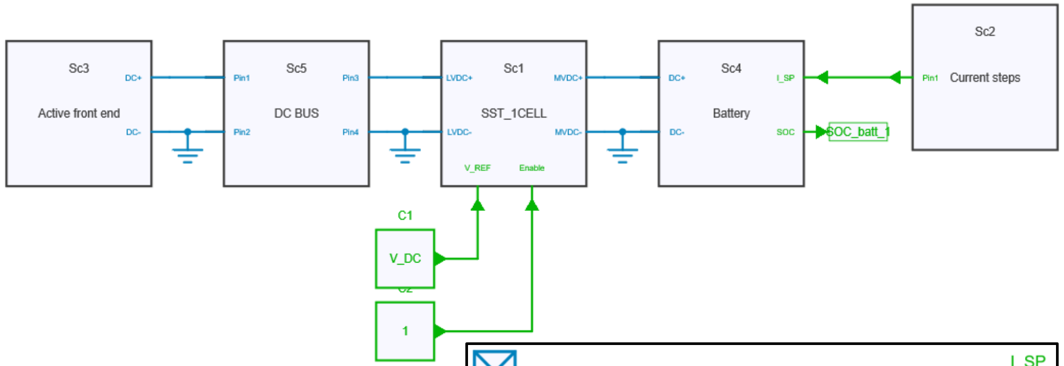
$$V_{AFE} = \frac{1}{C_{AFE}} \int (I_{FEED} + I_{AFE})$$

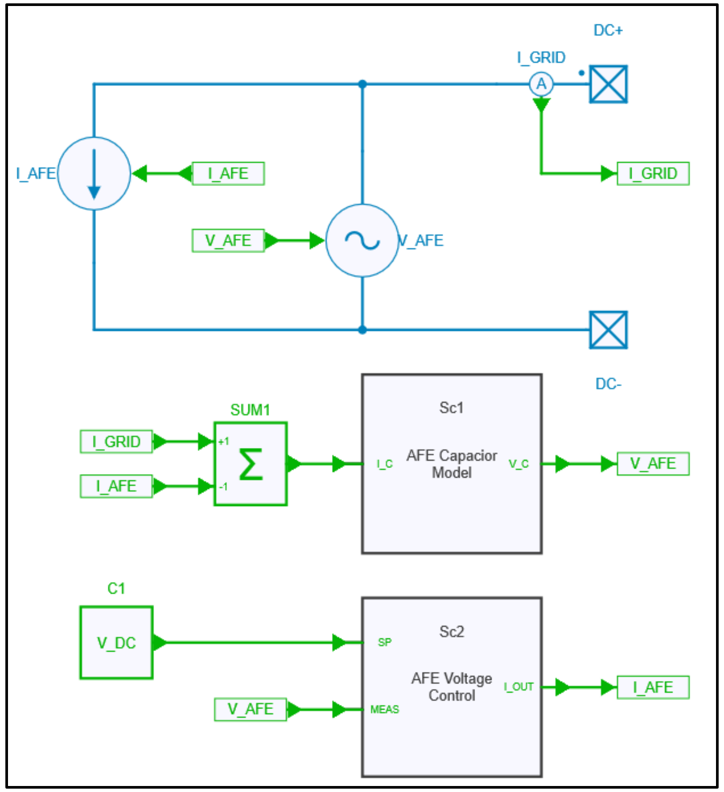
$$I_{FEED} = K_P(V_{AFE} - V_{DC}) + K_I \int (V_{AFE} - V_{DC})$$

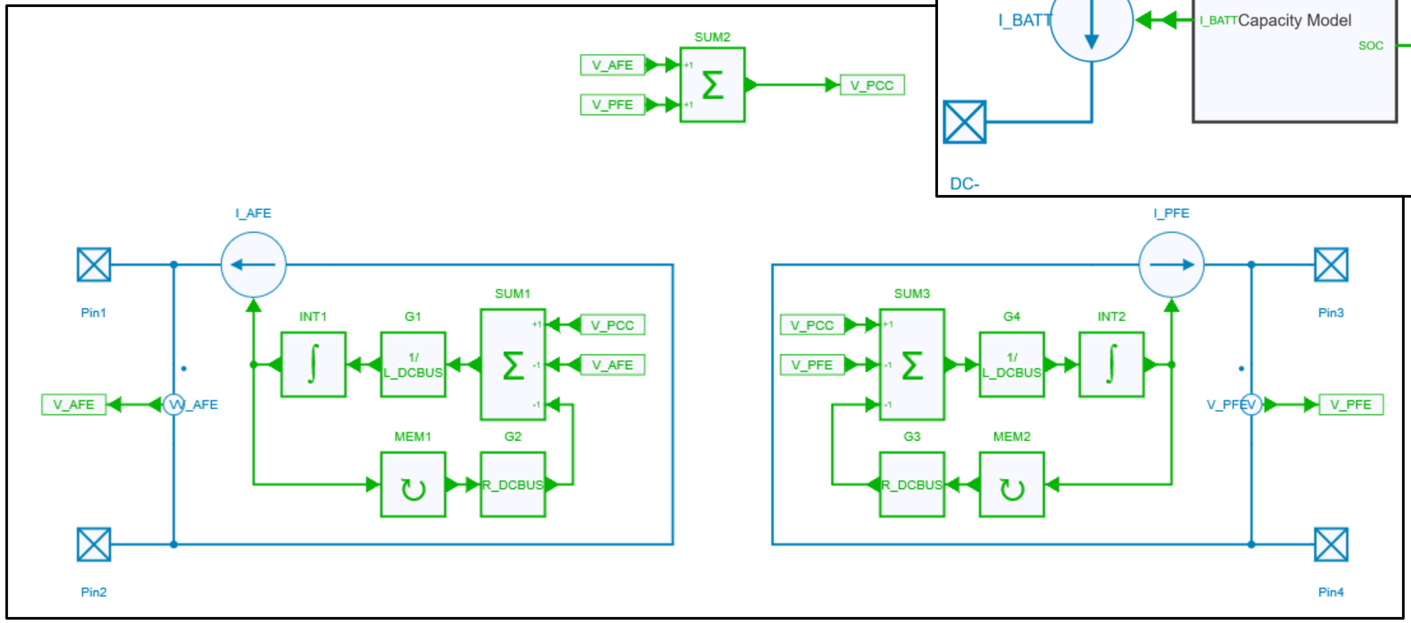
System level Models

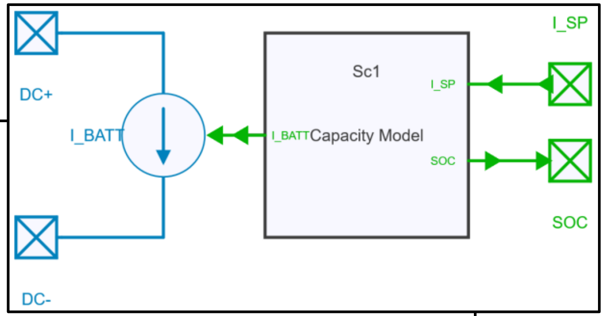
Active Front End and Battery Storage

- ❑ *Run model “3 Single SST with BESS and AFE from file “SST_DCMicroGrid_Models.jsimba”*



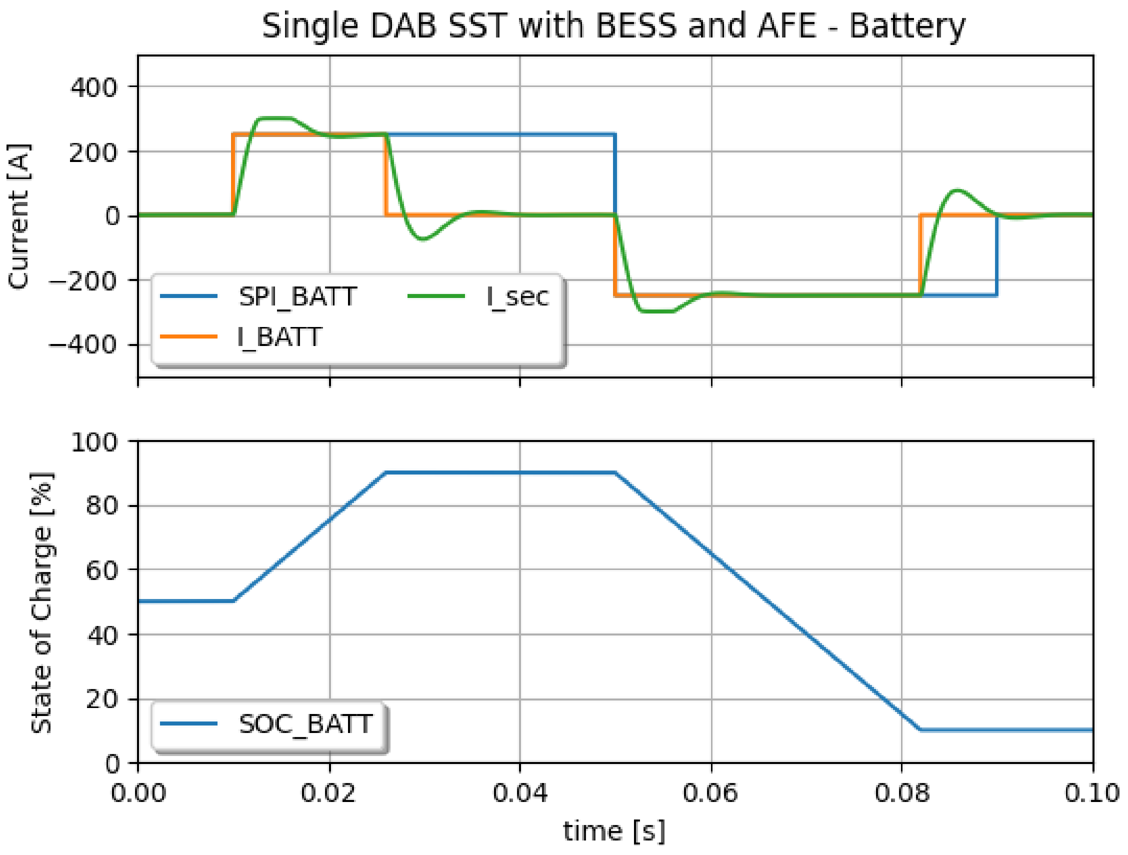


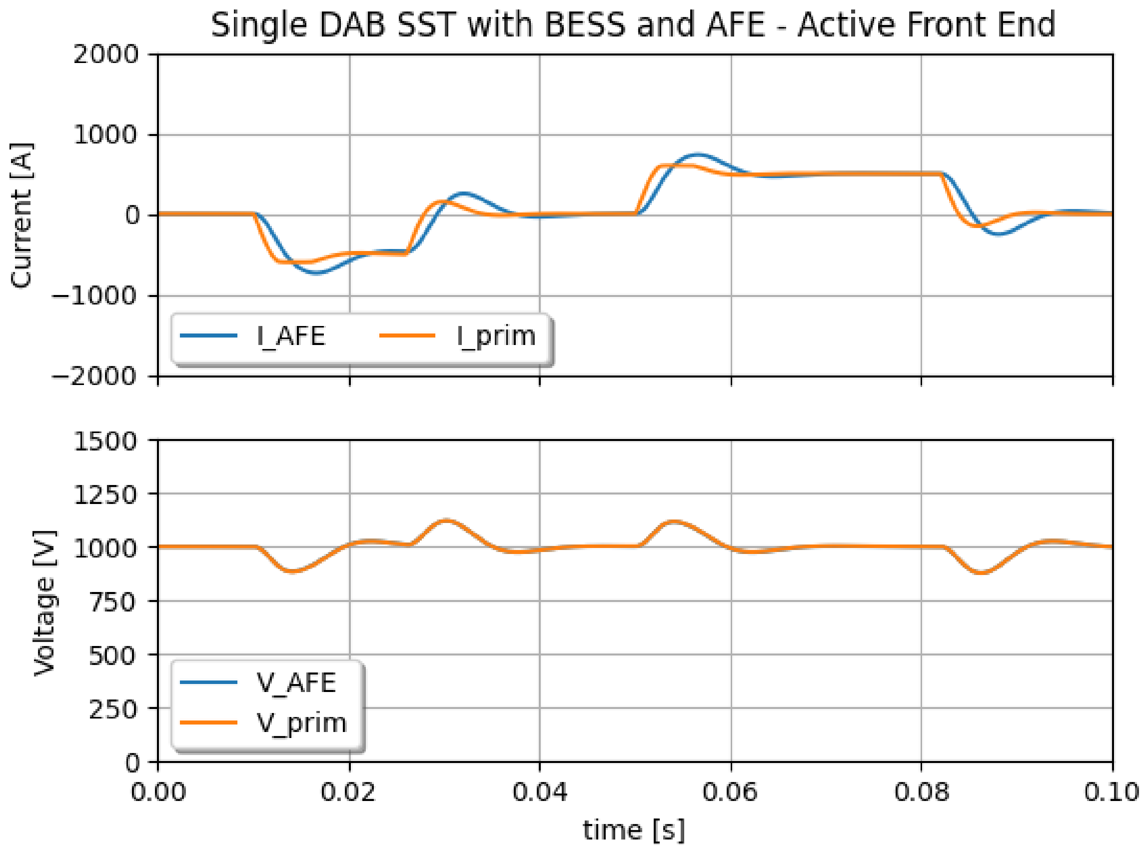




System level Models

Active Front End and Battery Storage





System Level Models

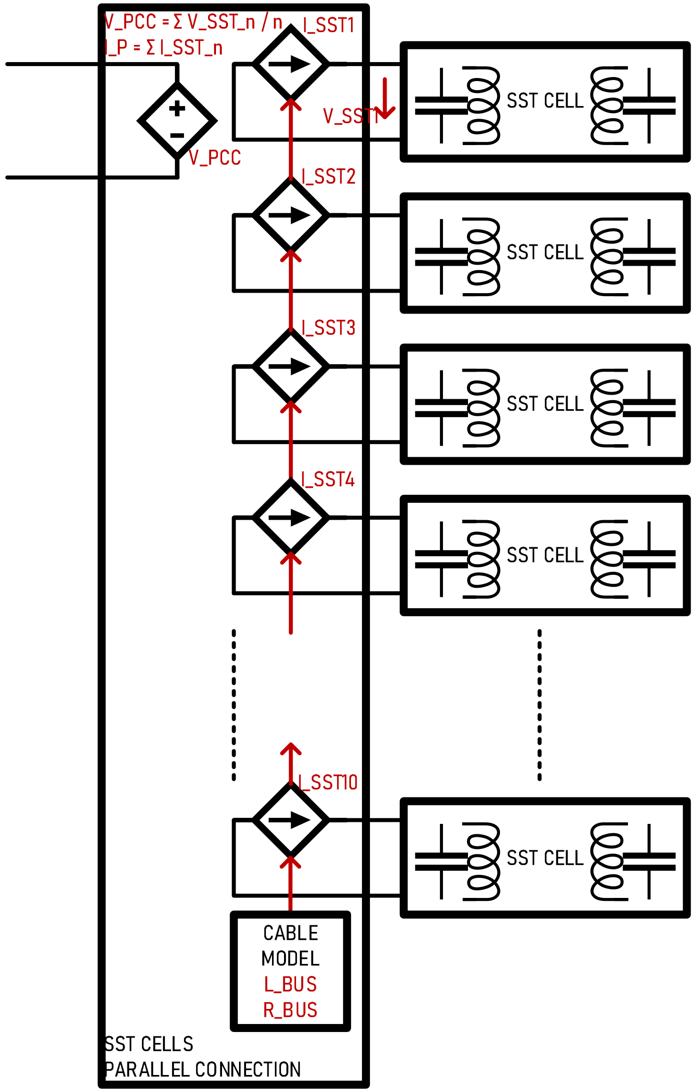
Series / Parallel connection of SST cells

- ❑ Connecting voltage source in series is not an issue
- ❑ Connecting voltage sources in parallel goes through current sources that implement a cable model
- ❑ The output voltage is a voltage source taking the mean value between them all

$$V_{PCC} = \sum \frac{V_{PRIM\_N}}{N}$$

$$I_{GRID} = \sum I_{PRIM\_N}$$

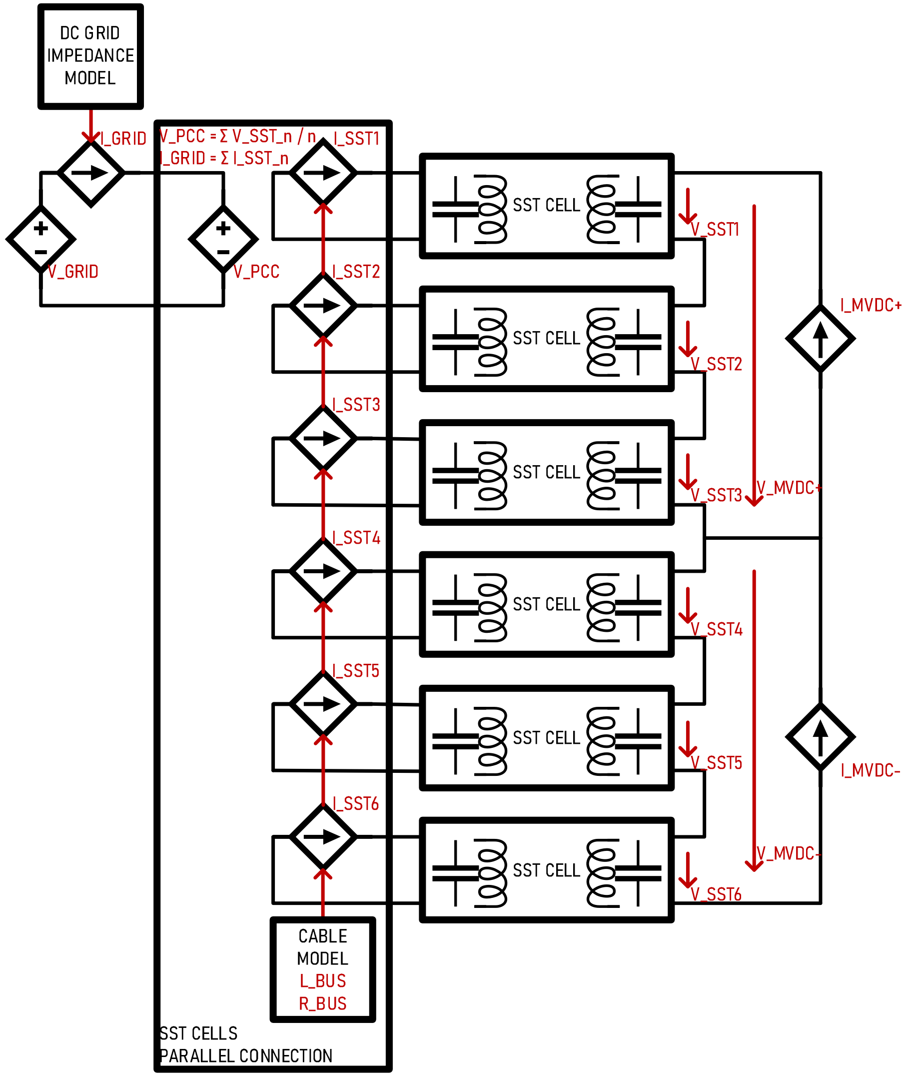
$$I_{SST\_N} = \frac{1}{L_{DCBUS}} \int (V_{PCC} - V_{PRIM\_N} - R_{DCBUS} I_{PRIM\_N})$$



SYSTEM LEVEL MODELS

Multicell IPOS configuration

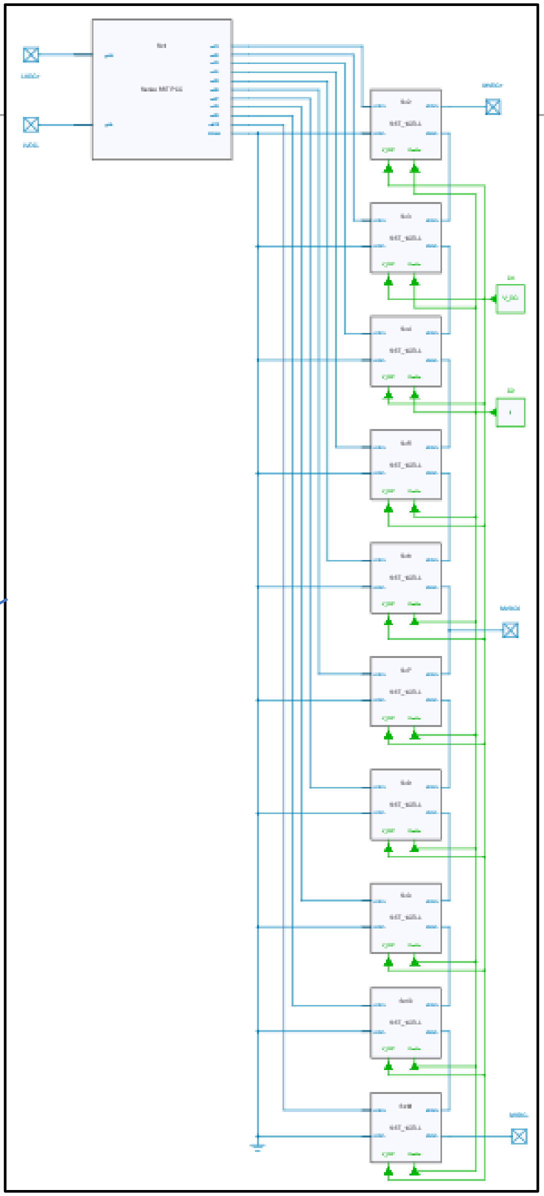
- The IPOS stands for Input Parallel Output Series
- The Output (right hand side) is connected to a three port MVDC with two current sources which may be asymmetric
- Voltages are controlled separately on each SST Cell secondary side
- LVDC Input (left hand side) implements a grid impedance model

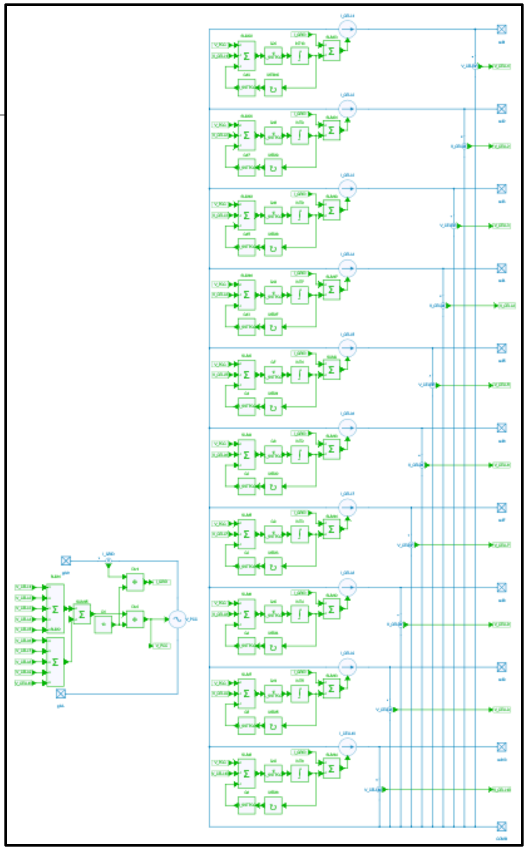


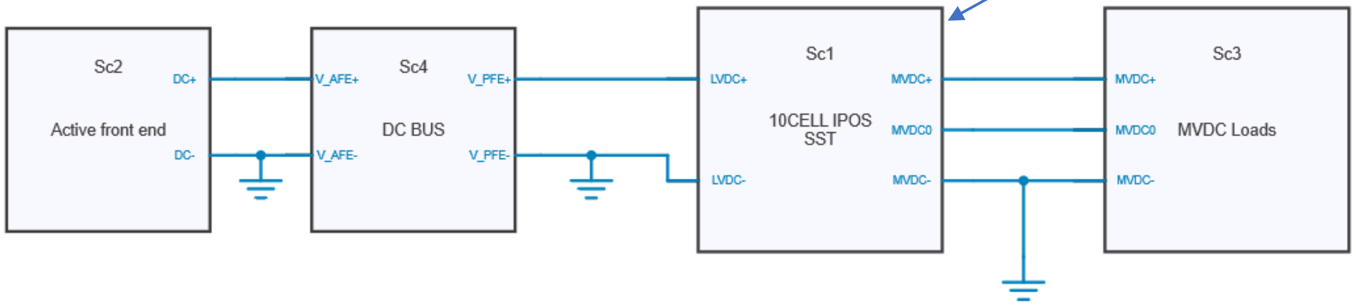
SYSTEM LEVEL MODELS

Multicell IPOS configuration

- ❑ *Run model “4 MultiCell IPOS SST”*
  from file “SST_DCMicroGrid_Models.jsimba”

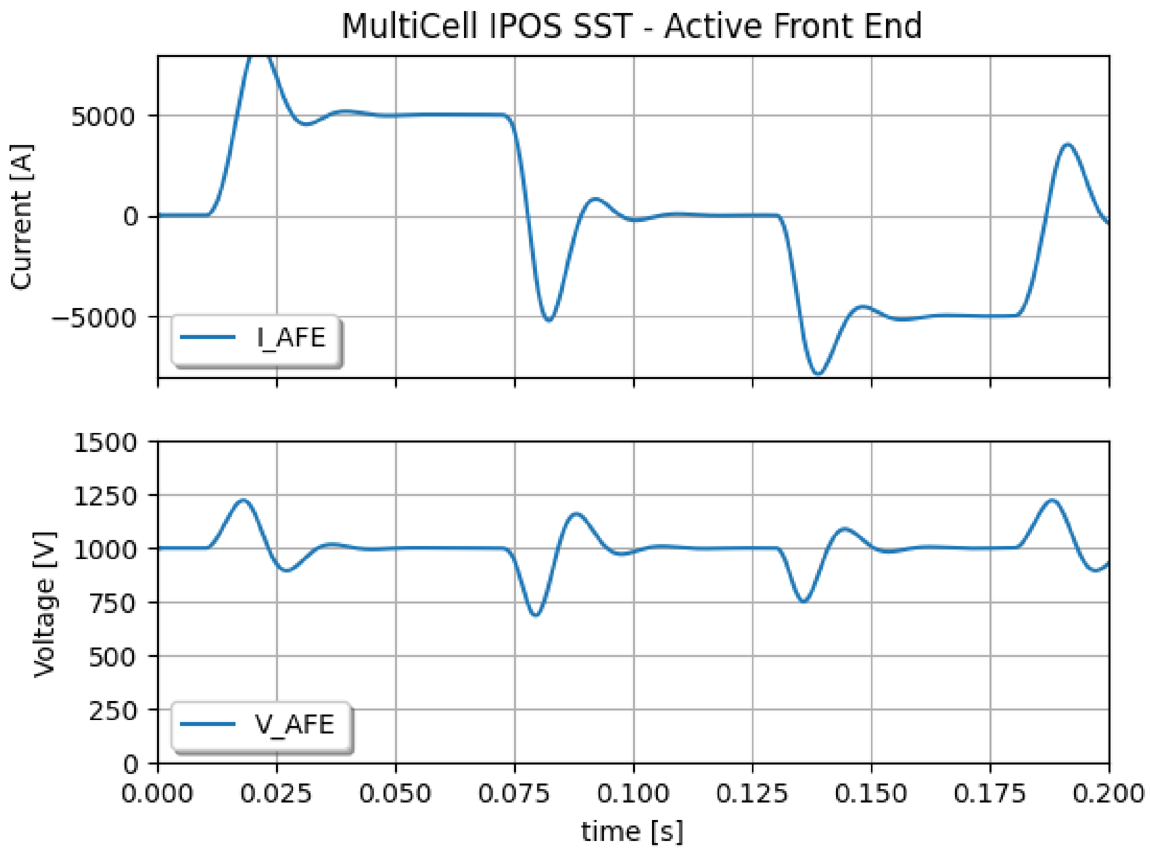


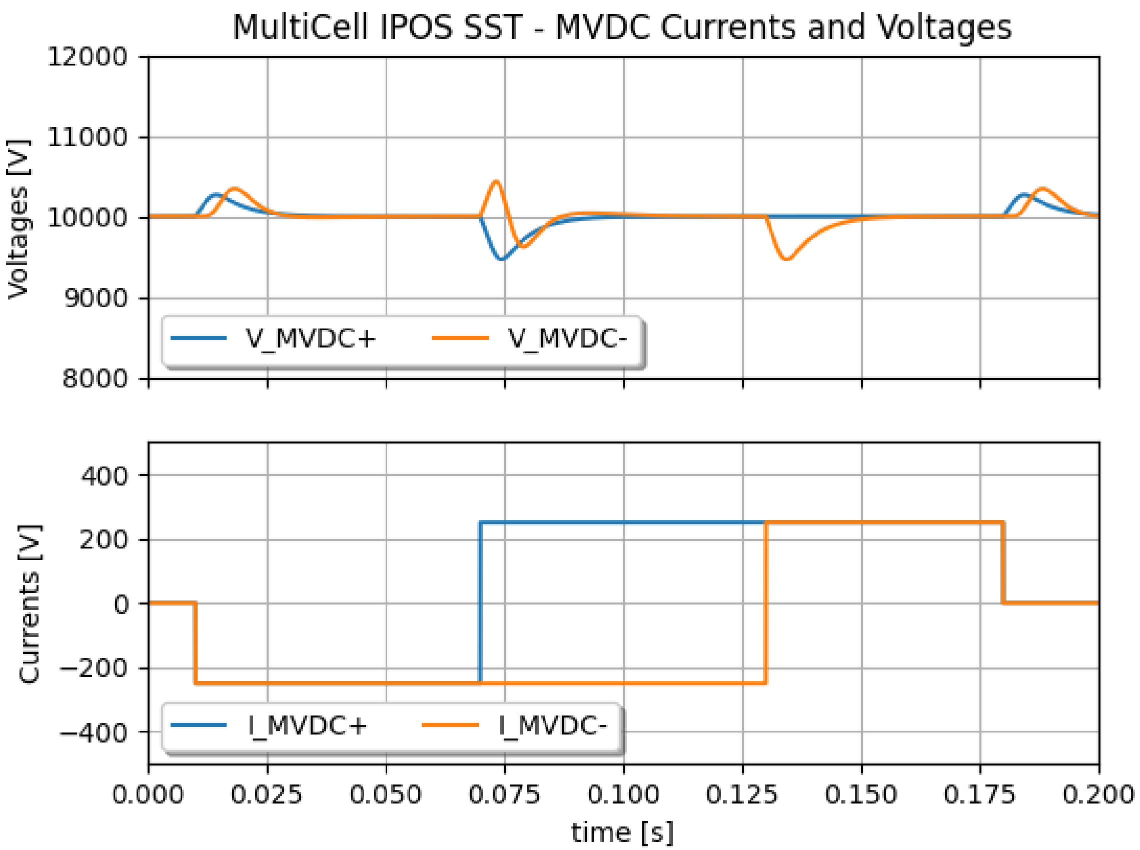




System level Models

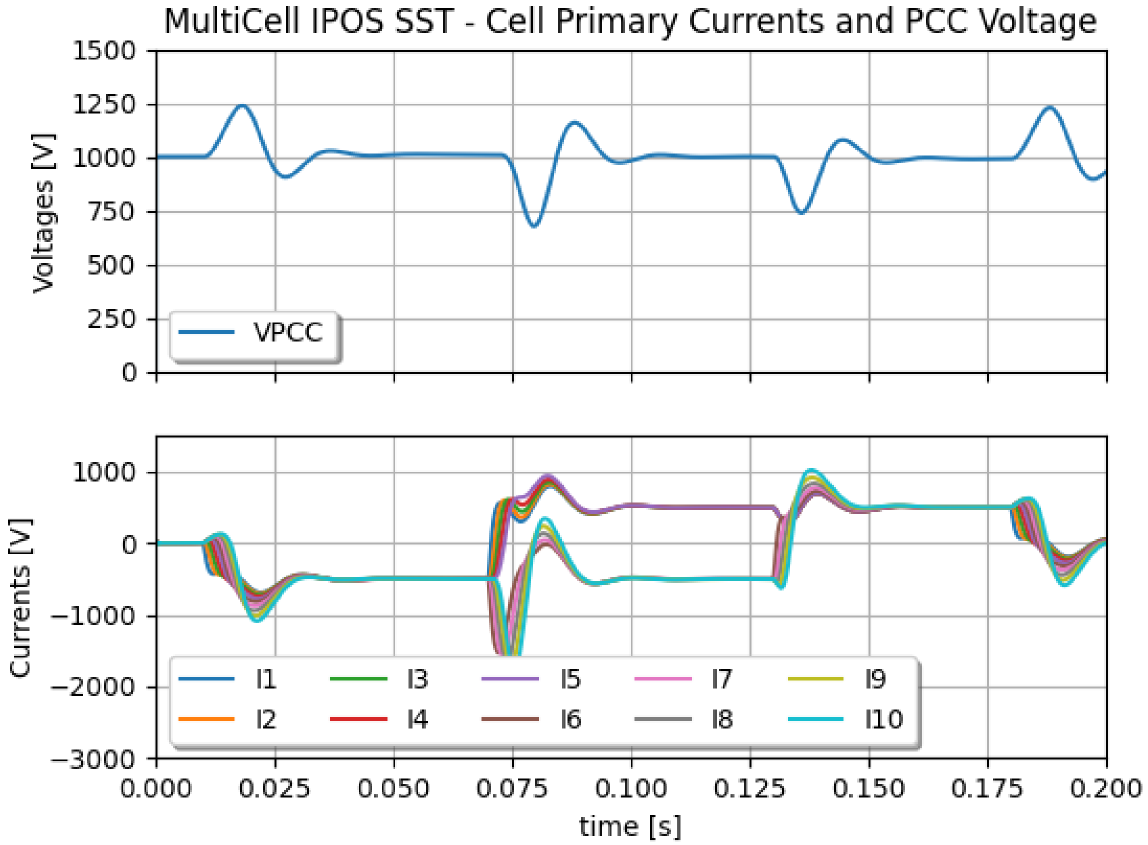
Multicell IPOS configuration

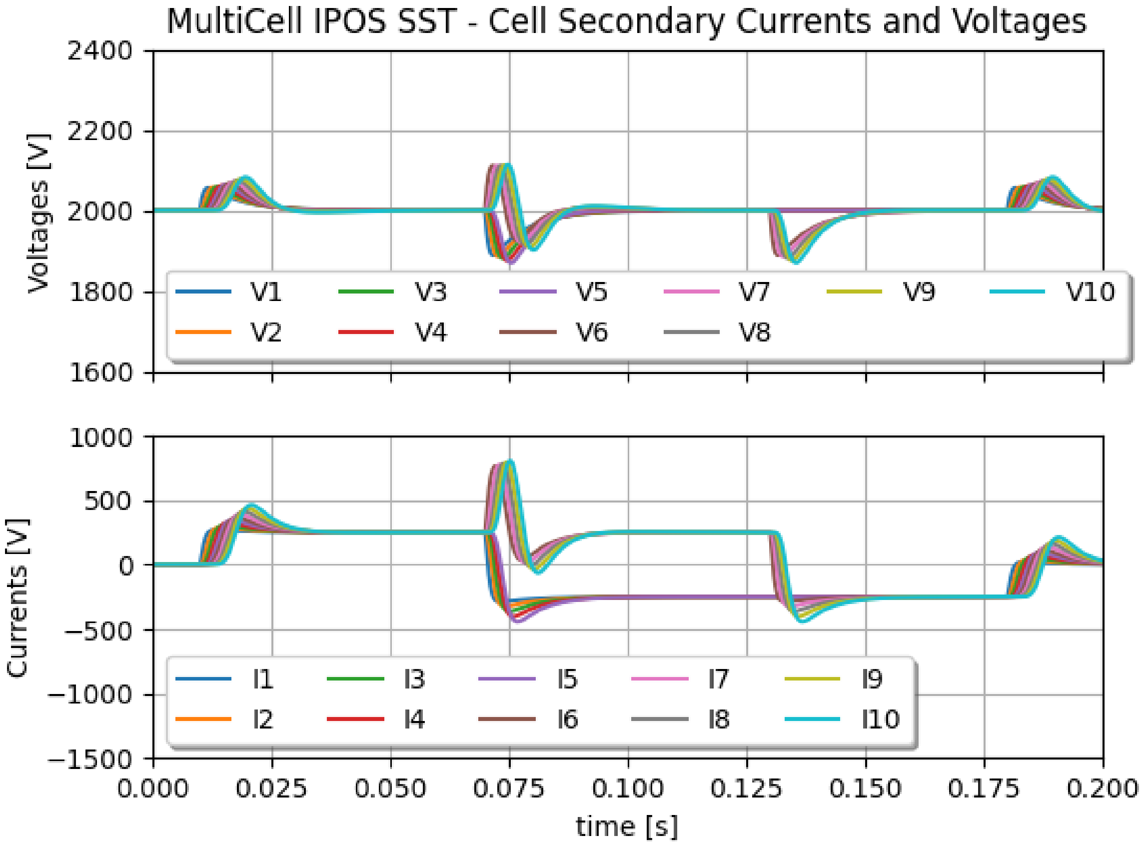




SYSTEM LEVEL MODELS

Multicell IPOS configuration

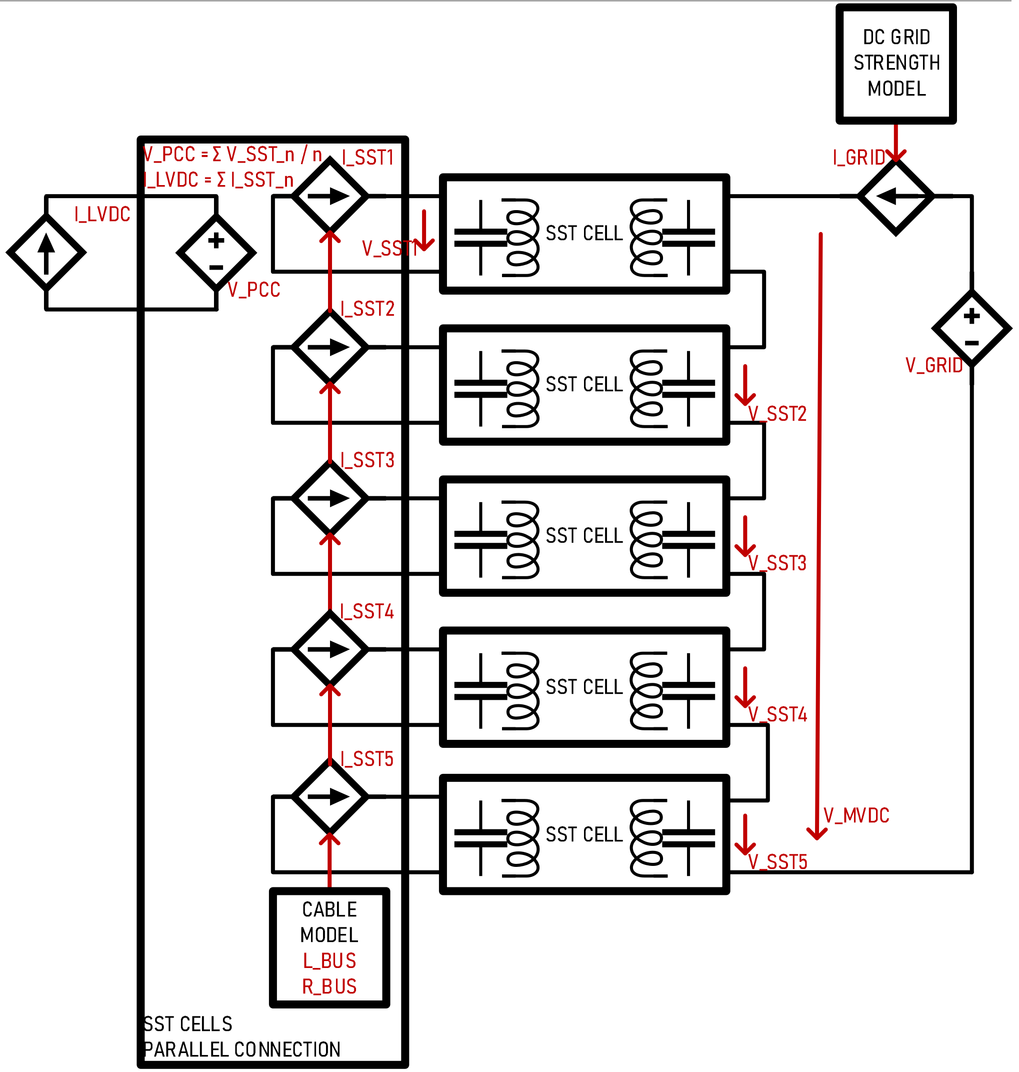




SYSTEM LEVEL MODELS

Multicell ISOP configuration

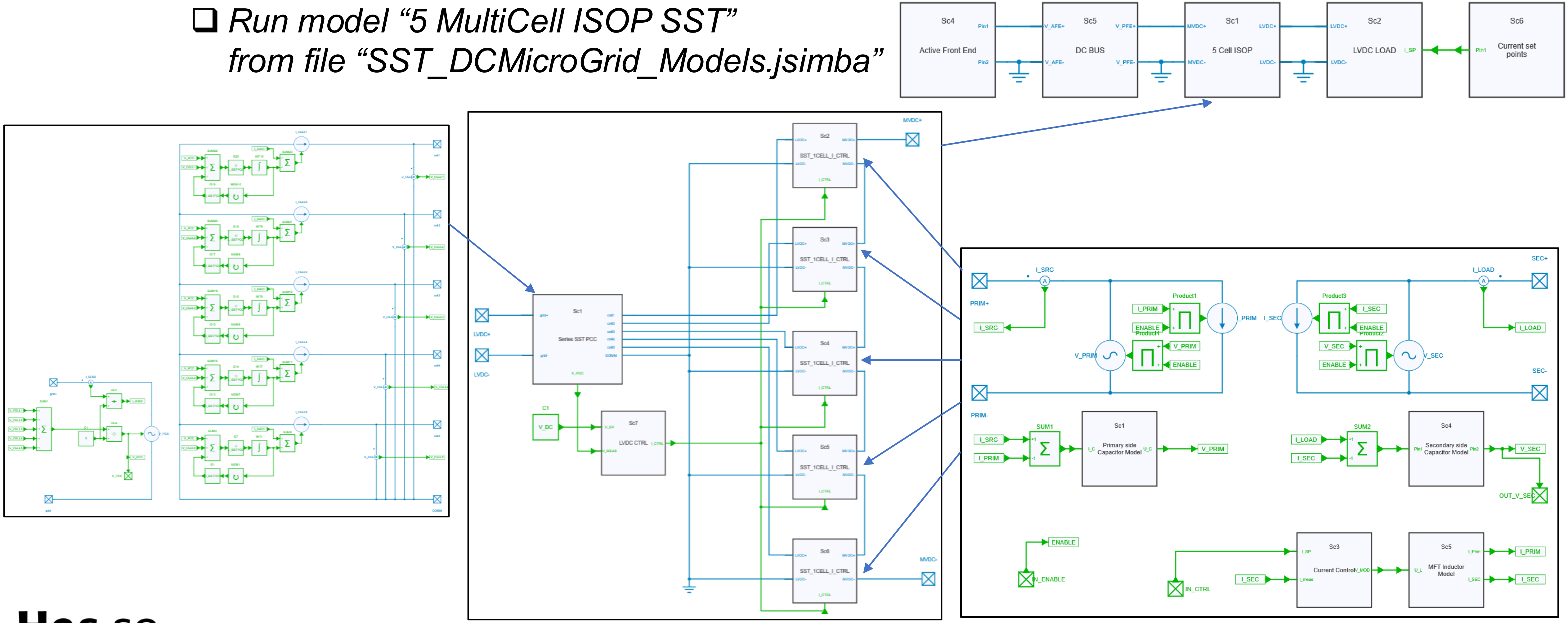
- The ISOP stands for Input Series Output Parallel
- The Output (left hand side) is connected to an LVDC current source
- Voltage is controlled on primary side by a single controller, each cell takes same current set point for ensuring secondary side voltage balance
- MVDC Input (right hand side) implements a grid impedance



SYSTEM LEVEL MODELS

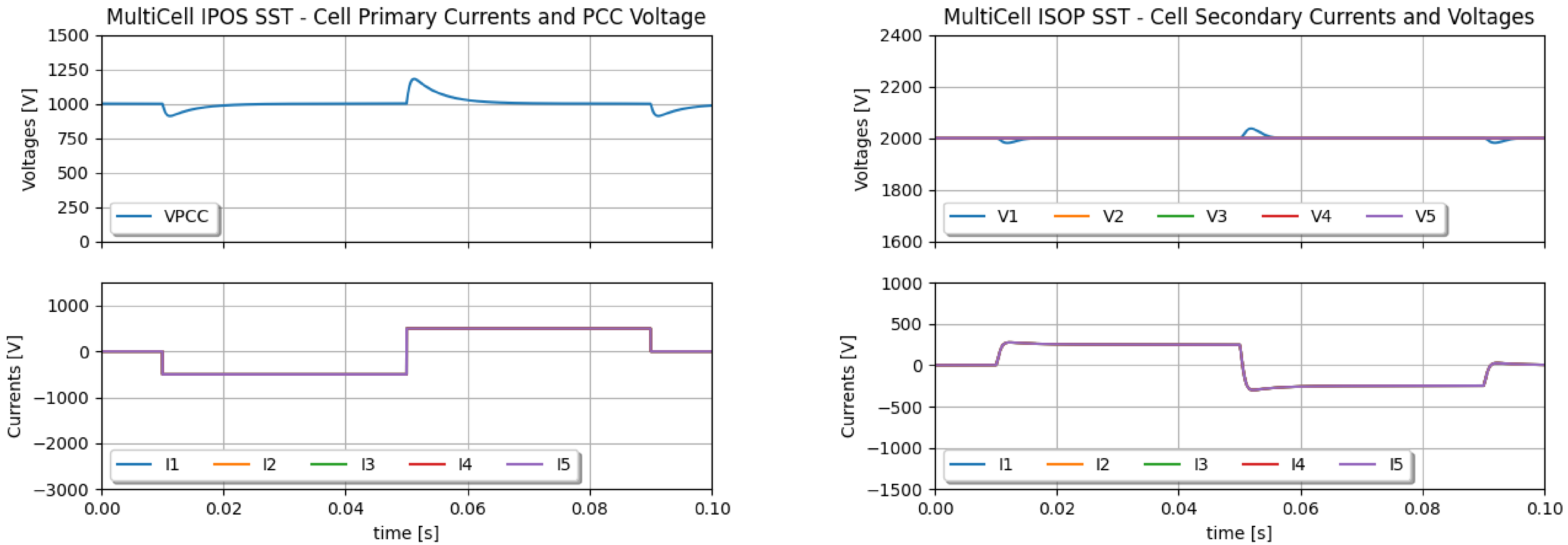
Multicell ISOP configuration

- ❑ *Run model "5 MultiCell ISOP SST" from file "SST_DCMicroGrid_Models.jsimba"*



SYSTEM LEVEL MODELS

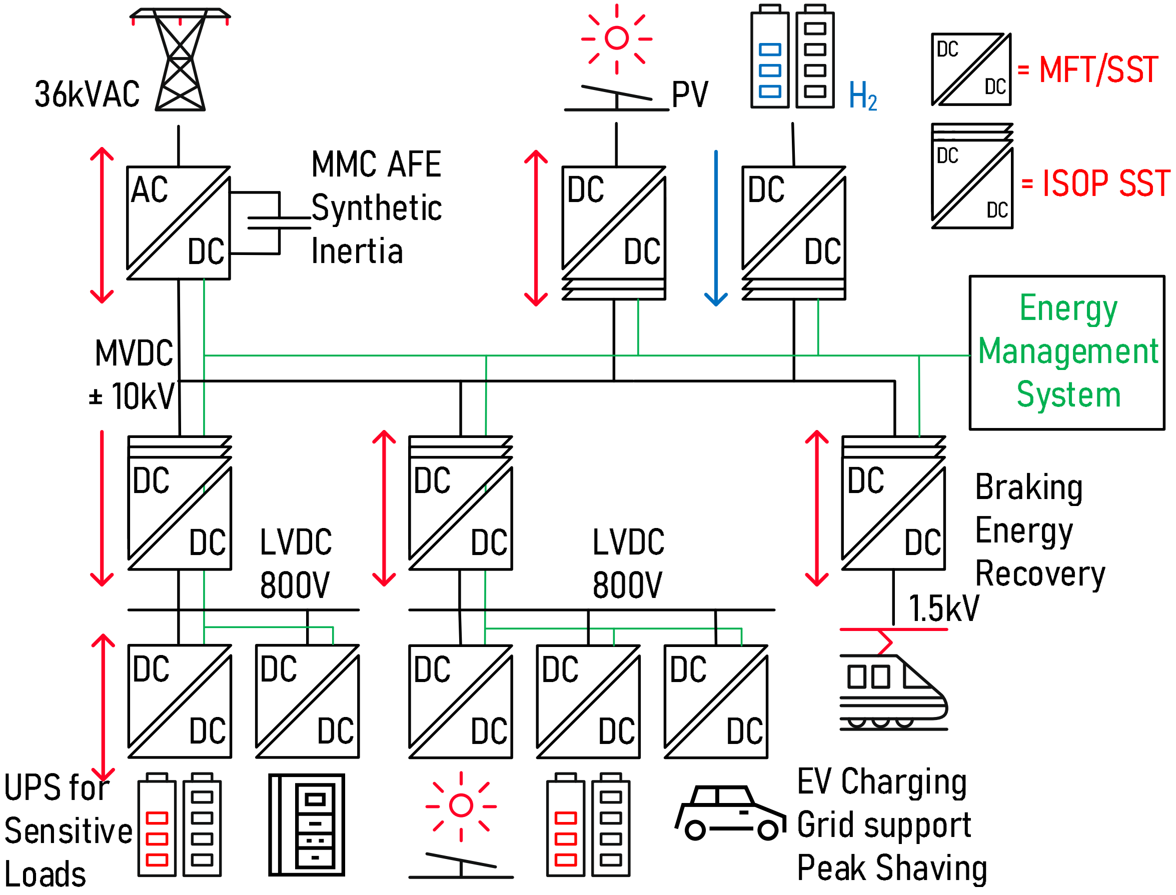
Multicell ISOP configuration



Model of an MVDC microgrid

Full model of an MVDC power system with multiple AFE/PFEs

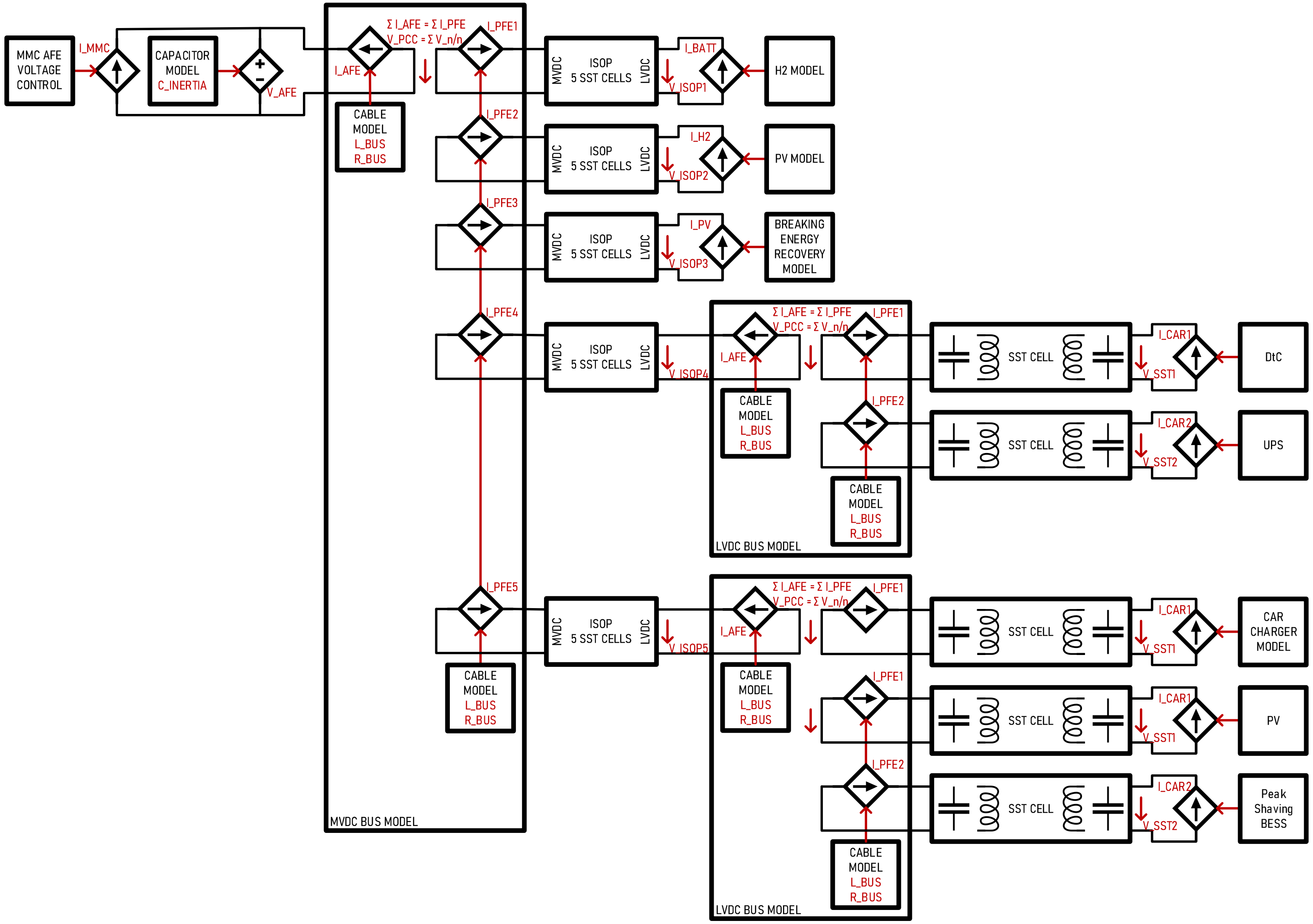
- ❑ In the following example, a full microgrid is operated with an MVDC backbone
- ❑ The MVDC backbone is in this case controlled by one active front end connected to the grid
- ❑ It features sources, loads, storage, and two connection to LVDC buses
- ❑ LVDC links features local battery storage for either UPS function or peak shaving



MODEL OF AN MVDC MICROGRID

Full model of an MVDC power system with multiple AFE/PFEs

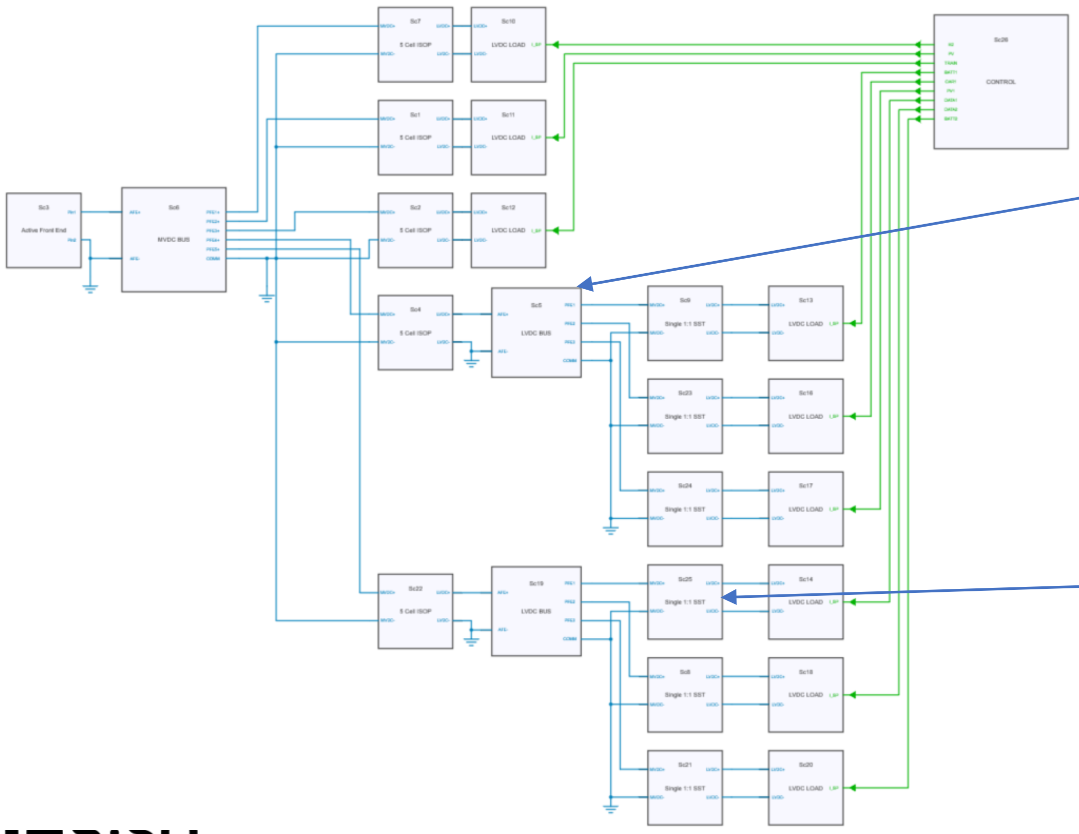
- ❑ The model is built with all previously presented elementary blocks
- ❑ About 30 SSTs are independently running in this model

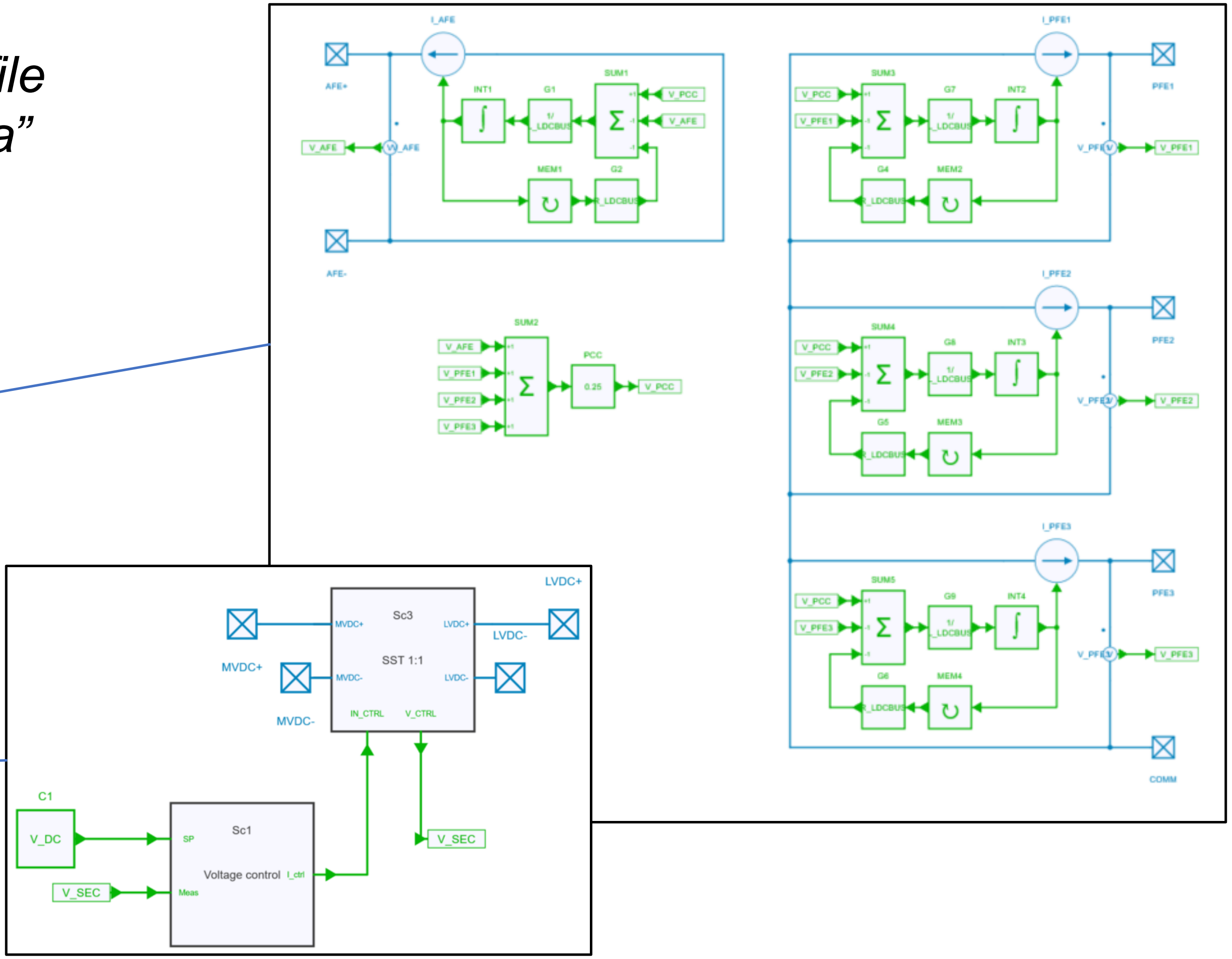


MODEL OF AN MVDC MICROGRID

Full model of an MVDC power system with multiple AFE/PFEs

- ❑ *Run model "6 DCMicroGrid" from file "SST_DCMicroGrid_Models.jsimba"*





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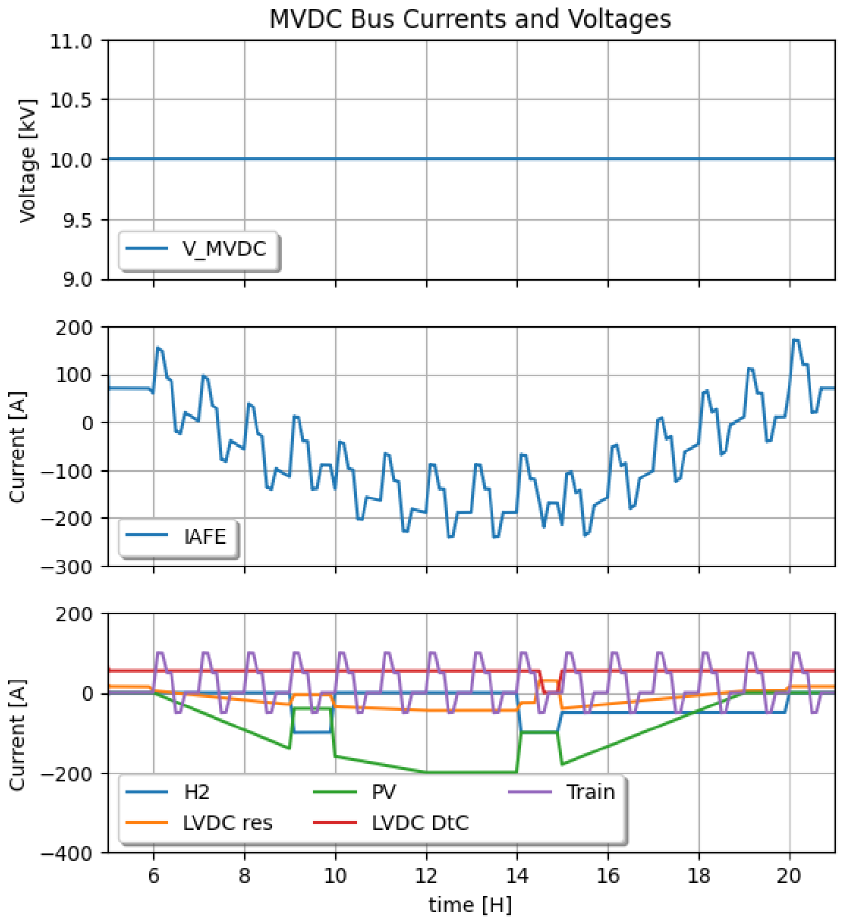
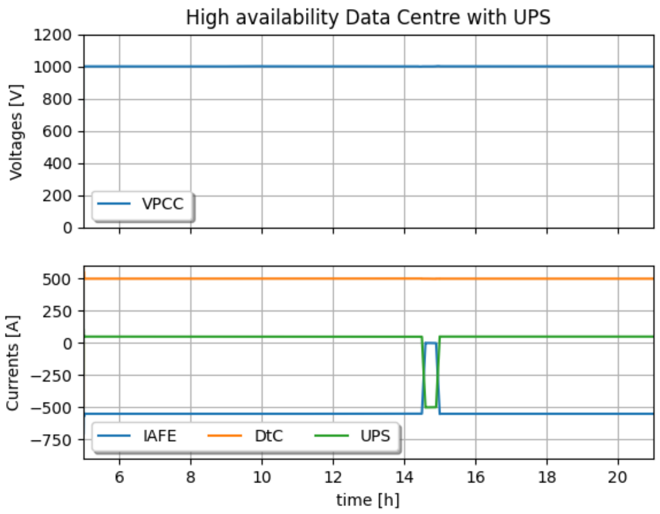
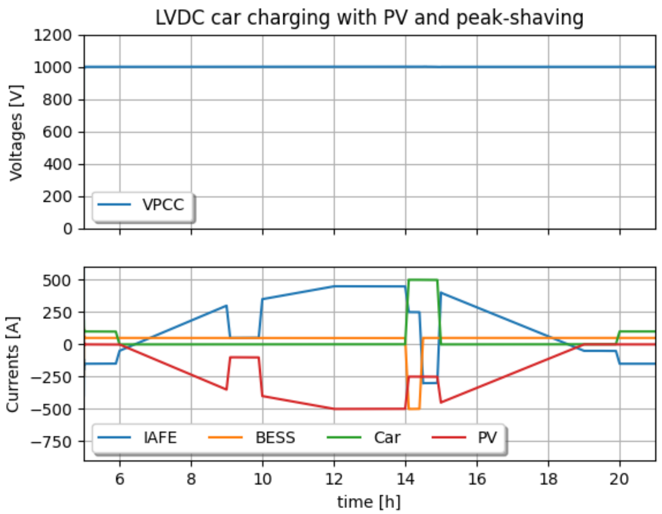
Model of an MVDC microgrid

Full model of an MVDC power system with multiple AFE/PFEs

- ❑ At some point, a car fast charge is requested, but a PV drop occurs
- ❑ Power is taken from utility grid but AFE is not enough for all loads
- ❑ DtC temporarily switches to UPS for saving the day



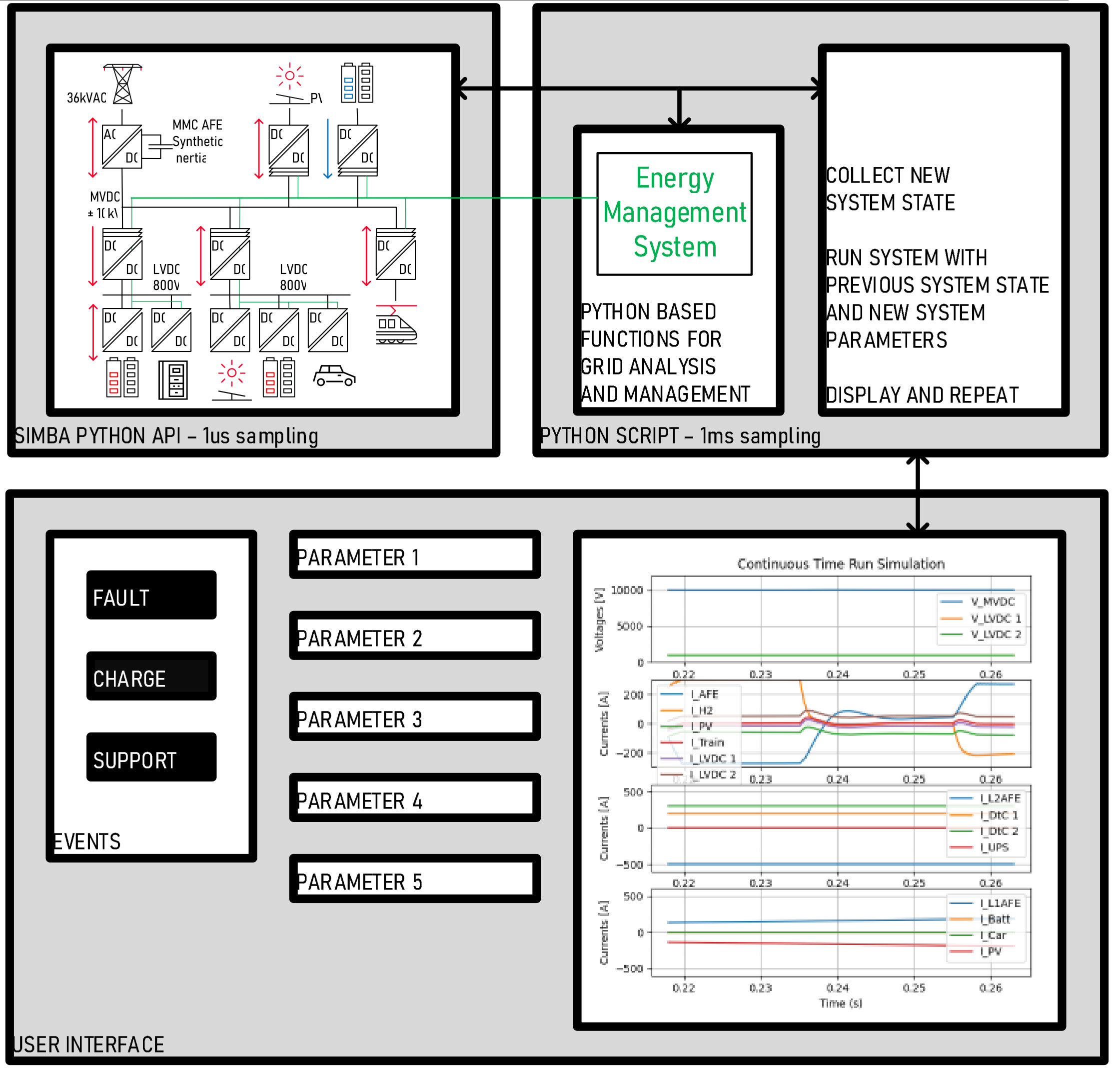




MODEL OF AN MVDC MICROGRID

Continuons run Approach

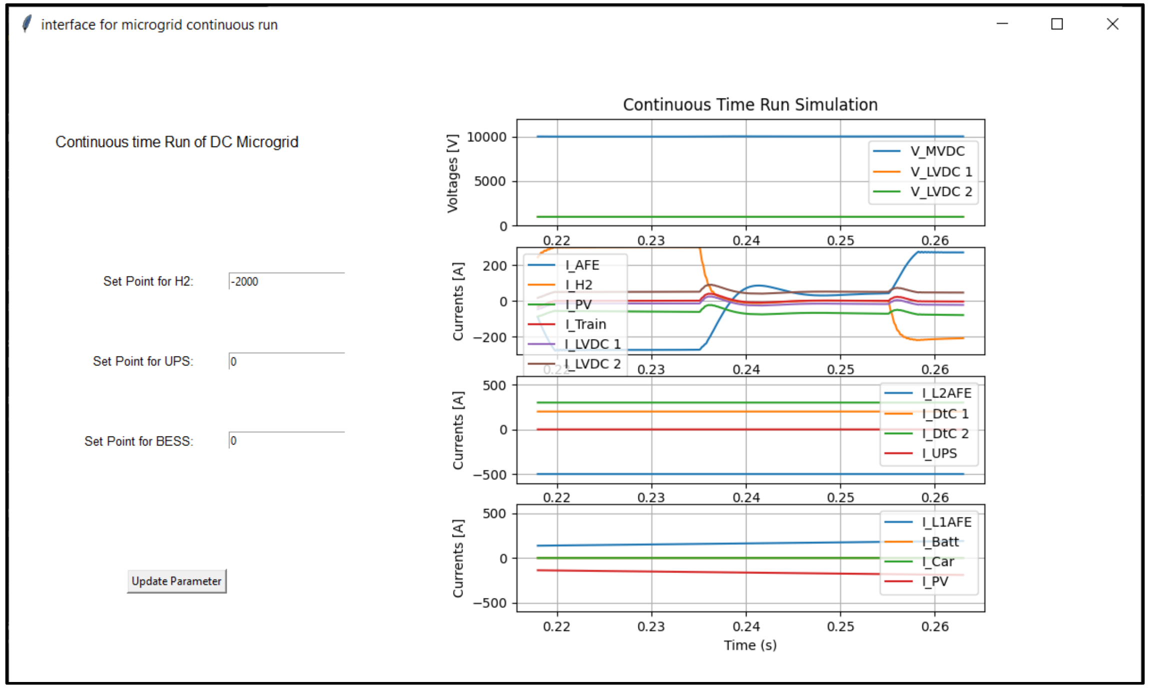
- ❑ The Simba model
  - Running at pseudo 1us sampling rate
  - Snapshot principle
- ❑ A Python user interface
  - Plot visualization
  - Parameters / set points settings
  - Possible alarms
- ❑ A Phython script
  - Calling the Simba Python API
  - Running at pseudo 1ms sampling rate
  - Implementing possible automated energy management features

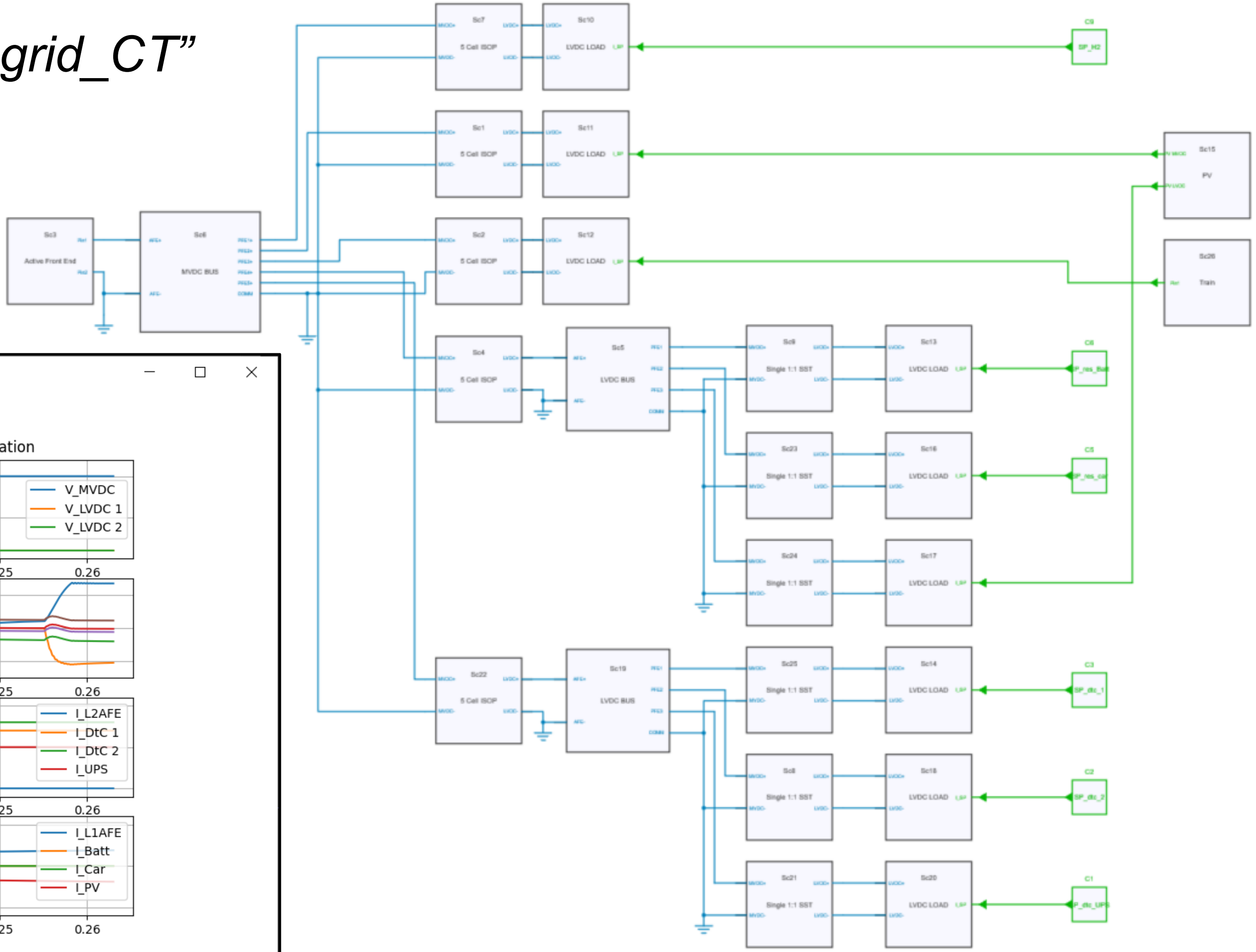


MODEL OF AN MVDC MICROGRID

Continuons run Approach

- ❑ *Run Python script “Run_7_microgrid_CT”*





CONCLUSIONS AND PROSPECTS

Contact me

- ❑ daniel.siemaszko@heig-vd.ch

What this tool is

- ❑ A didactic tool for simulating complex DC power systems with SST dynamics
- ❑ An approach to DC power system study
- ❑ Next steps would include
  - Modeling of droop controls for MVDC
  - Hardware and software protection schemes, short circuit current control
  - Possibly the implementation of any Python toolboxes applicable to control power systems

Get the models

- ❑ https://github.com/PESC-CH/System-level-MVDC-with-SST

https://simba.io/share/BOkZ3MTIOphfx2d0GO_r_5r0t1-WJD-FqdD6U_JANPuuzBj2vQq08uga4vFgK4WX/

Conclusions and Prospects

References

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